

# HOARDINGS360

## The Operating System for Outdoor Media

### MASTER PRODUCT DOCUMENT

Strategy · Product Requirements · Architecture · Release Plan · Go-to-Market

| Field          | Detail                                                                                                                      |
|----------------|-----------------------------------------------------------------------------------------------------------------------------|
| Document title | HOARDINGS360 — Master Product Document (Consolidated PRD, Roadmap & GTM Plan)                                               |
| Version        | v1.0 — Baseline for Sprint 0                                                                                                |
| Status         | For review and sign-off by founders, engineering leadership and commercial leadership                                       |
| Owner          | Product Owner, HOARDINGS360                                                                                                 |
| Audience       | Founders / Investors · Engineering Lead & Full-Stack Team · UI/UX Design · QA · Sales & Marketing · Operations & Onboarding |
| Supersedes     | OOH_Marketplace_PRD.docx and all prior concept notes, validation memos and module wish-lists                                |
| Review cadence | Re-baselined at the end of every release; reviewed monthly by the steering group                                            |

#### How to use this document

This is the single source of truth for HOARDINGS360. Sections 1–7 are for stakeholders and investors: why this business wins and how it makes money. Sections 8–15 are for engineering and design: architecture, domain model, module specifications and non-functional requirements. Sections 16–19 are the release plan and go-to-market plan that bind the two together. Sections 20–23 cover metrics, risk, governance and open decisions. **Nothing in this document may be built that is not traceable to a module ID (M01–M35) and a release version (V1.0–V5.0).**

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# 1. Executive Summary

HOARDINGS360 is a two-sided platform for India's out-of-home (OOH) advertising industry. It combines an **operational system of record for media owners** with a **public marketplace for advertisers**, connected by one shared inventory database.

The industry problem is not that billboards are hard to sell. It is that nobody knows, at any given moment, which billboards are free, which permits are about to lapse, which structures are damaged, and which client contracts are ending in three weeks. That information lives in spreadsheets, WhatsApp groups, printed permit files and the memory of a sales manager. Because it is invisible, inventory sits vacant, licences expire without warning, and the standard industry response to a finished campaign is to print a flex board with the owner's own phone number and hang it up — an advertisement for the absence of a system.

## 1.1 The strategy in one page

| Question                     | Our answer                                                                                                                                                                                                                              |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What are we building?        | An Outdoor Media Operating System (asset, compliance, contract, campaign and field-operations management for media owners) with a public marketplace layered on top of the same data.                                                   |
| Who do we sell to first?     | Media owners and OOH agencies operating 50–5,000 structures in Tier-1 and Tier-2 South India. Supply first, demand second.                                                                                                              |
| Why will they pay?           | Because a single missed municipal permit renewal or one 45-day vacancy costs more than a year of subscription. We sell risk removal and vacancy reduction, not "listings".                                                              |
| Why will advertisers come?   | Because we will be the only place in India where they can see verified, photographed, permit-valid inventory with real availability dates, on a map, without making twenty phone calls.                                                 |
| What is the defensible moat? | The digitised inventory graph: every board with GPS, photos, permit history, occupancy history and pricing history. Competitors can copy features; they cannot copy five years of asset history that owners themselves depend on daily. |
| How do we make money?        | Subscription (per-tenant + per-board slabs), transaction commission on marketplace bookings, featured listings, and enterprise/white-label licences.                                                                                    |
| What kills us?               | Launching a marketplace with no inventory. The first objective is not software — it is onboarding owners and digitising boards.                                                                                                         |

## 1.2 The product in three surfaces

| Surface                  | Working name | Primary user                                                | Core promise                                                                            |
|--------------------------|--------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Owner platform (web)     | H360 Manage  | Media owner, operations manager, sales executive, finance   | "Never lose a permit, never lose a day of occupancy, never lose a document."            |
| Public marketplace (web) | H360 Market  | Brands, agencies, SMEs, political and real-estate campaigns | "See every available board in the state, with real photos, real dates and real prices." |
| Field app (mobile / PWA) | H360 Field   | Installers, electricians, supervisors, inspectors           | "Scan the board, do the job, prove it with a geo-tagged photo."                         |

### 1.3 Release plan at a glance

| Version       | Theme                                 | Indicative window | The one thing it must prove                                                                          |
|---------------|---------------------------------------|-------------------|------------------------------------------------------------------------------------------------------|
| V0 (Sprint 0) | Architectural foundation              | Weeks 1–4         | The schema and event bus can absorb every future module without a rewrite.                           |
| V1.0          | Owner OS Core (private beta)          | Months 2–4        | 10 media owners run their permits and contracts on us daily and stop using their spreadsheet.        |
| V1.1          | Field, QR & Proof of Display          | Month 5           | Field teams close jobs in the app; clients accept our photo proof instead of phoning.                |
| V2.0          | Public Marketplace launch             | Months 6–8        | Advertisers find boards through search and send qualified enquiries that convert.                    |
| V2.1          | Media Plan Builder & CRM              | Month 9           | A sales rep produces a client-ready proposal in under 10 minutes instead of 3 hours.                 |
| V3.0          | Transact — quotes, bookings, payments | Months 10–12      | Money moves through the platform and GST-compliant invoices are issued automatically.                |
| V4.0          | Enterprise Operations                 | Months 13–16      | A 1,000-board agency runs incidents, work orders, landowner leases and true per-board P&L on us.     |
| V5.0          | Intelligence & Network                | Months 17–24      | The data itself becomes the product: pricing intelligence, vacancy prediction, APIs and white-label. |

Windows are indicative and assume the staffing model in Section 22. Scope is fixed per release; dates flex. **We will not compress a release by removing the architectural guardrails in Section 8.**

## 2. Market Opportunity & Validation

### 2.1 Market size — and an honest reading of it

Published estimates of the Indian OOH market vary widely because different analysts include or exclude transit, retail and place-based media. Stakeholders should treat the following as a range, not a number:

| Source / basis                                                 | Estimate                                                      | Note                                                                                                     |
|----------------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| WPP Media "This Year, Next Year" (Dec 2025)                    | Total India OOH ≈ ₹4,120 crore in 2026, growing ~8.6%         | Conservative; measures advertiser spend on classical OOH, with DOOH driving most of the growth.          |
| Industry trade estimates (AdGully / operator commentary, 2026) | ₹6,000–8,000 crore                                            | Broader definition including transit, retail and unorganised local inventory; higher growth assumptions. |
| Mordor Intelligence (2026)                                     | USD ~543 million (≈ ₹4,500 crore) in 2026, ~4.5% CAGR to 2031 | Narrow classical-OOH definition.                                                                         |
| DOOH sub-segment (multiple sources)                            | ₹1,500–2,000 crore by 2026; 15–20% of total OOH               | Fastest-growing segment; digitisation of premium inventory is the driver.                                |

#### The number that actually matters to us

Our revenue is not a share of ad spend — it is a share of **operator overhead and lost occupancy**. If India has roughly 100,000–150,000 organised large-format and digital structures under owners with 25+ boards, and average realised rent per board is ₹40,000–₹1,20,000 per month, then a 10–15% vacancy rate represents several hundred crore of recoverable annual revenue. Recovering even a fraction of it, plus removing permit-expiry penalties, comfortably funds a ₹1,500–₹15,000 per-month subscription. **Validate this bottom-up with the first 10 owners before scaling spend (see Section 23, OQ-1).**

### 2.2 Why now

- **Municipal digitisation.** Corporations across India are moving outdoor advertising policy, permits and site registers onto GIS-based systems. Owners who cannot produce a clean, document-backed compliance record are increasingly exposed.
- **Structural growth in sites.** Smart City programmes, national highway expansion, metro rail and airport modernisation are creating new OOH sites faster than owners' manual processes can track them.
- **DOOH normalises data expectations.** As digital screens bring impressions, scheduling and proof-of-play into OOH conversations, advertisers are beginning to demand the same accountability from static sites. Static owners without data will lose share.
- **Cheap, mature building blocks.** Geospatial databases, WhatsApp Business messaging, Aadhaar-based e-signature, UPI and object storage are all commodity infrastructure in India today. A platform that would have needed ₹10 crore in 2016 can reach production for a fraction of that now.
- **No entrenched Indian winner.** The market has software vendors and it has agencies, but no single player owns the operational record of the asset **and** the demand-side marketplace on top of it.

### 2.3 The two problem sets

#### 2.3.1 Media owner problems (our paying customer)

| Problem               | What it costs today                  | What HOARDINGS360 does                        |
|-----------------------|--------------------------------------|-----------------------------------------------|
| No live view of which | Boards stay vacant for weeks after a | Occupancy engine derives status from contract |

| Problem                                                     | What it costs today                                                                  | What HOARDINGS360 does                                                                                                            |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| boards are free                                             | campaign ends because nobody updated the sales team                                  | dates automatically; vacancy appears on a dashboard the day it is predictable, not the day it is noticed                          |
| Permits and NOCs expire unnoticed                           | Penalties, forced removal, structures sealed, revenue stopped mid-campaign           | Compliance engine tracks every clearance separately with its own authority, cycle and expiry, and escalates at 90/60/30/15/7 days |
| Documents scattered across email, drives and physical files | Hours lost per audit or dispute; missing evidence in legal proceedings               | Unified document vault; every document attached to any entity, versioned and auditable                                            |
| Field work coordinated by phone and WhatsApp                | No proof of installation, repeat site visits, disputes with clients over start dates | Work orders, technician assignment, GPS check-in, before/after photos, SLA tracking                                               |
| Selling requires manual PowerPoint decks                    | 3–4 hours per proposal, inconsistent pricing, slow response to briefs                | Media Plan Builder generates branded PDF/PPT and an interactive shareable link in minutes                                         |
| True profitability per board is unknown                     | Loss-making structures kept for years; landowner rent and electricity untracked      | Landowner lease and expense ledger produces net margin per board, per district                                                    |
| Vacant boards advertise the owner's own phone number        | A visible admission of unsold inventory; devalues the site and the brand             | Auto-publication to the marketplace 30–60 days before availability, while the current campaign is still live                      |

### 2.3.2 Advertiser problems (our demand side)

- **Discovery.** There is no single, trustworthy database of available structures with photographs, dimensions, illumination, traffic context and — critically — the date each becomes free.
- **Opacity.** Rates are relationship-driven and unpublished; the same site can be quoted at materially different prices to different buyers.
- **Planning speed.** A multi-city campaign requires contacting dozens of owners individually and reconciling inconsistent formats of information.
- **Verification.** After paying, the advertiser has no systematic proof that the creative went up on time, in the right place, lit at night, and stayed up for the full period.

## 2.4 Structural analogy

| Industry                    | Fragmented past                                           | Aggregated present                                           |
|-----------------------------|-----------------------------------------------------------|--------------------------------------------------------------|
| Hotels                      | Direct phone bookings, no visibility of availability      | Online travel marketplaces with live availability calendars  |
| Real estate                 | Local brokers, opaque pricing                             | National property portals with photographs and listed prices |
| Ride-hailing                | Street hailing, negotiated fares                          | Aggregator apps with transparent pricing and tracking        |
| Outdoor advertising (India) | Phone calls, spreadsheets, WhatsApp, printed permit files | <b>Currently unaggregated — this is the opening</b>          |

## 3. Vision, Positioning & Strategic Principles

### 3.1 Vision

#### Vision statement

Every outdoor advertising structure in India, digitised as a living asset — with a verified location, a valid permit, a known occupancy and a transparent price — and available to any advertiser in the country within minutes.

### 3.2 Mission (next 24 months)

Become the daily operating system for 250 media owners across South India, digitise 25,000 structures, and convert that inventory graph into the country's most trusted outdoor media marketplace.

### 3.3 Positioning statement

#### Positioning

For **media owners and OOH agencies managing 50 to 5,000 structures**, who currently run their business on spreadsheets, WhatsApp and physical files, **HOARDINGS360** is an **outdoor media operating system with a built-in marketplace** that turns every structure into a tracked, compliant, revenue-accountable asset — and automatically sells the empty ones. Unlike **listing directories** (which only advertise inventory) and **generic ERPs** (which know nothing about permits, flex installation or landowner leases), HOARDINGS360 is built around the physical hoarding as the core entity.

### 3.4 What we are deliberately NOT building (in the first 24 months)

| Not building                                            | Why                                                                                | Revisit at                  |
|---------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------|
| A DOOH content management / play-out system             | Mature global and Indian vendors exist; we integrate rather than compete           | V5.0 via API integration    |
| A programmatic DSP or ad exchange                       | Requires demand-side scale and measurement standards we will not have early        | V5.0 (supply-side API only) |
| An audience measurement panel                           | Capital-intensive; better sourced from partners and mobility data vendors          | V5.0 via partnership        |
| A creative design studio / flex artwork tool            | Not our problem to solve; distracts from the operating system                      | Not planned                 |
| A full accounting suite (ledgers, TDS, filings)         | We produce invoices and expenses and export to Tally/Zoho — we do not replace them | Integration only, V3.0      |
| A consumer-facing "advertise your shop" self-serve tool | Small-ticket demand, high support cost, distracts from agency and brand demand     | Evaluate post-V3.0          |

### 3.5 Six strategic principles

1. **Supply before demand.** A marketplace with 200 boards is a liability; it teaches advertisers that we are empty. We do not open the public marketplace until we clear the inventory threshold defined in Section 16 (V2.0 entry criteria).
2. **The ERP is the wedge; the marketplace is the prize.** Owners adopt us because we solve their permit and vacancy pain. That adoption is what fills the marketplace. Never invert the order.



3. **The board is the centre of the universe.** Every module, every screen, every report resolves back to a board. The Board 360 page (Section 11, M02) is the most important screen in the product.
4. **Nothing is ever deleted.** Assets, agreements, permits and incidents are append-only with full history. Outdoor media is a legally contested business; the audit trail is a feature, not overhead.
5. **No hardcoded business rules.** Incident categories, board types, status transitions, alert thresholds and document types are configuration, not code. Every agency operates differently.
6. **Earn the right to charge commission.** We start with subscription and lead generation. We only introduce transaction fees once we demonstrably deliver bookings the owner would not otherwise have won.

### 3.6 The core insight to protect

**Do not build this as a software product. Build it as a network.**

The technology is not the hard part. Inventory acquisition is. Every roadmap decision, every hire and every rupee of marketing spend in the first 12 months should be tested against one question: **does this get more verified boards onto the platform, or keep the ones we have?** Features that do neither can wait.

## 4. Users, Segments & Personas

### 4.1 Supply-side segmentation

| Tier               | Profile                                                                     | Boards       | Primary pain                                                                         | Product fit                                | Priority                                              |
|--------------------|-----------------------------------------------------------------------------|--------------|--------------------------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------|
| A — Enterprise     | National / large regional media groups, transit and airport concessionaires | 1,000–5,000+ | Field operations, compliance at scale, per-asset profitability, multi-branch control | Full ERP + white-label; long sales cycle   | V4.0 target; pilot 1–2 early for credibility          |
| B — Core (our ICP) | Established regional owners and OOH agencies, 2–5 cities                    | 50–1,000     | Permit expiry, vacancy, proposals, field coordination                                | Perfect fit for V1.0–V3.0                  | <b>Primary target from day one</b>                    |
| C — Long tail      | Single-city owners, family businesses                                       | 5–50         | Finding buyers; low process pain                                                     | Marketplace listing + light inventory tool | Free/low tier from V2.0 — supply density, not revenue |

#### Why Tier B is the beachhead

Tier C has pain we cannot charge for. Tier A has budget but 9–18 month procurement cycles and demands features we will not have until V4.0. Tier B is large enough to feel operational pain daily, small enough to decide in a single meeting, and — decisively — owns the inventory density that makes a regional marketplace credible.

### 4.2 Personas

#### Persona 1 — Rajesh, Managing Director (Media owner, 220 boards, 3 districts)

- **Goals:** maximise occupancy; avoid municipal trouble; know what the business actually earns per site.
- **Frustrations:** finds out about a vacancy when a client mentions it; discovers an expired permit when a notice arrives; cannot answer "which of my boards lost money last year?"
- **Buying trigger:** a penalty, a sealed structure, or a competitor who responded to a brief faster.
- **What wins him:** a compliance risk dashboard in the first demo, showing his own boards. **Success = he opens the dashboard every morning.**

#### Persona 2 — Priya, Operations Manager

- **Goals:** every campaign up on the start date; every complaint closed the same week; every technician accounted for.
- **Frustrations:** coordinating 14 installers over WhatsApp; no record of who did what; clients calling to ask whether the flex is up.
- **What wins her:** work orders with GPS check-in and automatic proof-of-display delivery to the client. **Success = her WhatsApp group volume drops.**

#### Persona 3 — Arun, Sales Executive

- **Goals:** respond to briefs fast; never quote a board that is already booked; hit monthly targets.
- **Frustrations:** three hours in PowerPoint per proposal; stale availability data; chasing renewals from memory.
- **What wins him:** select 20 boards on a map, click once, get a branded deck and a shareable client link. **Success = proposal turnaround under 30 minutes.**

#### Persona 4 — Meera, Media Planner at an agency (Advertiser side)

- **Goals:** build a multi-city plan for a client brief within 48 hours, at a defensible price, with proof of execution afterwards.
- **Frustrations:** twenty phone calls, inconsistent rate cards, photographs that are three years old, no post-campaign evidence.
- **What wins her:** a map with verified availability dates and a shortlist she can export. **Success = she starts a plan on H360 instead of a spreadsheet.**

#### Persona 5 — Suresh, Field Technician

- **Goals:** know today's jobs, get there, finish, get paid.
- **Frustrations:** unclear instructions, wrong site addresses, being blamed for delays he cannot disprove.
- **What wins him:** one screen with today's tasks, offline-tolerant, a QR scan and two photos to close a job. **Success = jobs closed in-app without a supervisor call.**

#### Persona 6 — Deepa, Compliance/Legal Officer (Tier A)

- **Goals:** zero expired clearances; produce any document within a minute of a municipal query.
- **What wins her:** the compliance matrix and the document vault with version history and audit trail.

### 4.3 Role catalogue (RBAC — see Section 13.3 for the permission matrix)

| Role                         | Scope                                                                               | Introduced |
|------------------------------|-------------------------------------------------------------------------------------|------------|
| Platform Super Admin         | All tenants; verification, moderation, billing, platform analytics                  | V1.0       |
| Platform Support Agent       | Read-mostly cross-tenant access for support, with impersonation logging             | V2.0       |
| Tenant Owner / Company Admin | Full control of one tenant, including user and subscription management              | V1.0       |
| Operations Manager           | Assets, incidents, work orders, field staff, proof of display                       | V1.0       |
| Sales Executive              | Assigned clients, proposals, enquiries, bookings; sees rate cards but not cost data | V1.0       |
| Finance Officer              | Invoices, payments, landowner payouts, expenses, profitability reports              | V3.0       |
| Compliance Officer           | Permits, NOCs, certificates, document vault, compliance dashboard                   | V1.0       |
| Field Supervisor             | Assign and verify work orders in a geography                                        | V1.1       |
| Field Technician             | Own tasks only; mobile app; capture and close                                       | V1.1       |
| Advertiser (registered)      | Marketplace search, shortlist, enquiry, campaign tracking                           | V2.0       |
| Advertiser Sub-user          | Agency team member under an advertiser org                                          | V3.0       |
| Landowner (read-only portal) | View lease, payouts and their structure — optional, drives stickiness               | V4.0       |

## 5. Business Model & Pricing

### 5.1 Revenue streams

| #  | Stream                             | Mechanic                                                                      | Live from                                       | Role                                                       |
|----|------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------|
| R1 | SaaS subscription (media owners)   | Platform fee + per-board slab, billed annually or monthly                     | V1.0 (paid from V2.0; free during private beta) | <b>Primary, predictable revenue</b>                        |
| R2 | Transaction commission             | 5% of campaign value on marketplace-originated bookings, taken on payment     | V3.0                                            | Scales with GMV; must be earned                            |
| R3 | Featured / priority listings       | Owners promote inventory in search results and on the map                     | V2.1                                            | High-margin, low-effort                                    |
| R4 | Lead fee (alternative to R2)       | Fixed fee per qualified, accepted enquiry for owners not on transaction plans | V2.0                                            | Bridges the gap until payments exist                       |
| R5 | Enterprise & white-label           | Annual licence with SSO, custom domain, dedicated support, optional on-prem   | V5.0                                            | Large-ticket, Tier A                                       |
| R6 | Data & intelligence products       | Market rate benchmarks, occupancy indices, availability APIs for agencies     | V5.0                                            | Long-term moat monetisation                                |
| R7 | Onboarding & digitisation services | Paid field survey, photography and data capture for large owners              | V1.0                                            | Cost recovery; <b>not a profit centre</b> — it buys supply |

## 5.2 Recommended pricing architecture

Flat monthly tiers under-charge large owners and over-charge small ones. Use a **platform fee plus per-board slab** so price scales with the value delivered (boards under management), while keeping a free tier to build marketplace density.

| Plan                          | Boards    | Indicative price / month      | Includes                                                                                                                                |
|-------------------------------|-----------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Listing (Free)                | Up to 10  | ₹0                            | Marketplace listing, basic inventory record, enquiry inbox. No compliance engine, no alerts, no exports.                                |
| Starter                       | 11–75     | ₹2,499 + ₹25/board above 25   | Full asset registry, permit & agreement tracking, alerts (email + in-app), document vault, occupancy dashboard, 3 users.                |
| Professional                  | 76–400    | ₹5,999 + ₹20/board above 75   | Everything in Starter, plus WhatsApp alerts, field app, proof of display, incidents, media plan builder, CRM, 15 users.                 |
| Enterprise                    | 401–2,000 | ₹19,999 + ₹15/board above 400 | Everything in Professional, plus work orders, landowner leases, per-board P&L, vendor management, BI, SSO, unlimited users, API access. |
| Enterprise Plus / White-label | 2,000+    | Custom annual contract        | Custom domain and branding, dedicated environment option, priority SLA, implementation manager.                                         |

### Pricing discipline for the sales team

Never discount the list price — discount the **term** (annual prepay) or bundle **digitisation services**. Price cuts on a young SaaS product are irreversible and signal that the software is optional. The Founding Partner Program (Section 19.2) is the only sanctioned discount, it is time-boxed, and it is exchanged for something we need: inventory, a testimonial and a reference call.

## 5.3 Commission model (from V3.0)

| Booking origin                                                    | Commission                     | Rationale                                                                |
|-------------------------------------------------------------------|--------------------------------|--------------------------------------------------------------------------|
| Marketplace-originated (advertiser found the board on H360)       | 5% of campaign value           | We created the demand; standard aggregator economics                     |
| Owner-originated, contracted through H360                         | 0% (subscription only)         | We did not create the demand; charging would push contracts off-platform |
| Featured-listing-originated                                       | 5% + featured fee already paid | Owner opted into promotion                                               |
| Repeat booking of the same advertiser-board pair within 12 months | 2.5%                           | Rewards retention, discourages disintermediation                         |

### Disintermediation is the structural risk of every marketplace

Advertisers and owners will meet on H360 and try to transact outside it. Do not fight this with contracts alone — fight it with **utility**: contract generation, e-signature, GST invoicing, proof of display and payment protection that make transacting on-platform strictly easier. Instrument leakage from V3.0 (Section 20) and treat a rising leakage rate as a product failure, not a sales failure.

## 5.4 Illustrative unit economics (to be validated, not assumed)

| Metric                             | Working assumption     | Basis / note                                     |
|------------------------------------|------------------------|--------------------------------------------------|
| ARPA — Tier B owner (Professional) | ₹7,500–₹12,500 / month | Professional base plus per-board uplift, at 150– |

| Metric                            | Working assumption      | Basis / note                                                           |
|-----------------------------------|-------------------------|------------------------------------------------------------------------|
|                                   |                         | 400 boards                                                             |
| Annual contract value (Tier B)    | ₹0.8–1.3 lakh           | Assumes annual prepay at 2 months free                                 |
| CAC — Tier B (field-led sales)    | ₹35,000–₹60,000         | Includes digitisation cost of first 100 boards                         |
| Payback period                    | 4–6 months              | Acceptable for field-sold vertical SaaS                                |
| Gross margin (SaaS)               | 78–85%                  | After hosting, maps, messaging and storage                             |
| Annual logo churn target          | < 10%                   | Operational systems of record churn low once permit data lives in them |
| Net revenue retention target      | > 115%                  | Driven by board growth within accounts and plan upgrades               |
| Marketplace take rate at maturity | 4–5% of on-platform GMV | Blended across commission tiers                                        |

**Instruction to Finance/Founders:** replace every figure in this table with measured values by the end of V2.0. A business plan built on assumed CAC is the most common cause of over-hiring in Indian B2B SaaS.

## 6. Competitive Landscape & Differentiation

### 6.1 The four categories we compete with

| Category                                 | Examples                                                  | What they do well                                                                               | Where they leave a gap                                                                                                                               |
|------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| OOH management software (India)          | Edge1 and similar vendors                                 | Established inventory, campaign and billing tooling; DOOH CMS capability; enterprise references | Software-first, not network-first; limited public demand generation on behalf of the owner; compliance and landowner economics are typically shallow |
| OOH management software (global)         | ApparatiX, BillboardPlanet, Quattro, SignDash, Broadsign  | Mature contracting, lease, vinyl and scheduling workflows                                       | Built for US/EU permitting and leasing norms; no Indian municipal permit model, no WhatsApp, no GST, no rupee pricing, no local support              |
| Media buying agencies & planning portals | The Media Ant, Excellent Publicity and regional agencies  | Real demand, real client relationships, planning expertise                                      | They are buyers, not systems of record; they will never give owners operational tooling, and owners are wary of handing them inventory data          |
| Listing marketplaces / directories       | Mera Hoardings, Adboard Booking, various regional portals | Low-friction listing; some booking flow                                                         | Directory-only: no permit engine, no field operations, no proof of display; owners have no reason to keep the data fresh, so listings decay          |
| The real incumbent                       | Microsoft Excel + WhatsApp + a physical file cabinet      | Free, flexible, universally understood, zero training                                           | No alerts, no audit trail, no multi-user control, no demand generation — <b>this is who we must actually beat</b>                                    |

### 6.2 Our four differentiators

- 7. Compliance depth.** We model each clearance (municipal licence, traffic police NOC, structural stability certificate, electrical safety clearance, landowner permission) as a separate record with its own authority, renewal cycle and expiry. Competitors treat "licence" as a single date field. This is the feature owners feel most acutely.
- 8. The vacancy loop.** Contract end date drives automatic marketplace publication ("Available from 1 October") **while the current campaign is still running**, so the site is sold before it is empty. This single loop is our most quotable value proposition and should anchor all marketing.
- 9. Operations, not just inventory.** Incidents, work orders, technician dispatch, GPS check-in and photographic proof make us the system people open every day — not once a quarter when a listing changes.
- 10. True per-board profitability.** Landowner rent and electricity cost tracked against client revenue. No competitor in the Indian mid-market answers "which boards actually make money?"

### 6.3 Positioning map

Two axes matter: **operational depth** (does the system run the business day-to-day?) and **demand generation** (does it bring the owner buyers?). Directories score high on the second and near-zero on the first; management software is the reverse; spreadsheets score zero on both. HOARDINGS360 is the only entrant deliberately targeting the top-right quadrant, and the sequencing in Section 16 is designed to get there via operational depth first — because that is the axis owners will pay for before we have demand to offer.

## 6.4 Competitive response we should expect

- An existing software vendor bolts a public marketplace onto its product. **Counter:** our compliance and landowner depth, and the fact that their marketplace inherits only their own customers' inventory.
- A large agency launches an owner portal. **Counter:** owners will not give operational and pricing data to a buyer. Emphasise our neutrality — we never take a media-owner position or buy inventory ourselves. **This neutrality must be a written, published policy.**
- A well-funded directory buys inventory listings with free tiers. **Counter:** listings decay without an operational reason to keep them fresh. Compete on data freshness, not listing count — and publish a "last verified" date on every listing.



## 7. Product Scope Overview & Module Map

The complete product is defined as **35 modules across three surfaces**. Every module has a stable ID that is used in the backlog, in commit messages, in QA test plans and in the release matrix (Section 17). Modules are specified in Sections 11–13.

### 7.1 Master module map

#### Surface A — H360 Manage (Media owner platform, web)

| ID  | Module                                  | One-line purpose                                                    | Release |
|-----|-----------------------------------------|---------------------------------------------------------------------|---------|
| M01 | Identity, Tenancy & Access Control      | Multi-tenant isolation, users, roles, permissions, SSO              | V1.0    |
| M02 | Asset Registry & Board 360              | The digital twin of every structure; the centre of the system       | V1.0    |
| M03 | Document Vault                          | One polymorphic repository for every document on every entity       | V1.0    |
| M04 | Compliance & Permit Engine              | Multiple independent clearances per board, each with its own expiry | V1.0    |
| M05 | Client & Agreement Management           | Commercial contracts, versions, values, renewal tracking            | V1.0    |
| M06 | Availability & Occupancy Engine         | Derives live status and future availability from contract dates     | V1.0    |
| M07 | Notification & Alerting                 | Escalating multi-channel alerts driven by platform events           | V1.0    |
| M08 | Activity Feed & Audit Trail             | Append-only history for every entity; legal defensibility           | V1.0    |
| M09 | Owner Dashboard & Operational Reporting | Occupancy, revenue, risk and vacancy views                          | V1.0    |
| M10 | Bulk Import & Data Onboarding           | CSV/Excel import, validation, de-duplication, guided onboarding     | V1.0    |
| M11 | Incident Management                     | Configurable incident types, severity, lifecycle, resolution proof  | V1.1    |
| M12 | Proof of Display                        | Geo-tagged, timestamped installation evidence packaged for clients  | V1.1    |
| M13 | Campaign Management                     | Campaign as first-class entity spanning many boards                 | V2.1    |
| M14 | CRM & Sales Pipeline                    | Clients, contacts, communication log, renewal follow-ups            | V2.1    |
| M15 | Media Plan & Proposal Builder           | Board selection to branded PDF/PPT and shareable client link        | V2.1    |
| M16 | Quotation, Booking & Contract Lifecycle | Quote → hold → booking → agreement → e-signature                    | V3.0    |
| M17 | Invoicing, Payments & GST               | Tax-compliant invoices, payment collection, receipts, ageing        | V3.0    |
| M18 | Work Order & Field Force Management     | Task generation, dispatch, SLA, technician performance              | V4.0    |
| M19 | Landowner Lease & Expense Ledger        | Land rent, electricity and operating costs per structure            | V4.0    |
| M20 | Vendor & Procurement                    | Printers, fabricators, electricians; POs and performance scores     | V4.0    |
| M21 | Board Health Score                      | Composite asset health from compliance, condition and occupancy     | V4.0    |

| ID  | Module                                 | One-line purpose                                     | Release |
|-----|----------------------------------------|------------------------------------------------------|---------|
| M22 | Financial Intelligence & Profitability | Revenue less cost per board, district and client     | V4.0    |
| M23 | Business Intelligence & Heatmaps       | Cross-tenant-safe analytics, geographic and temporal | V5.0    |
| M24 | AI Pricing & Vacancy Prediction        | Rate recommendations and churn/vacancy forecasting   | V5.0    |

### Surface B — H360 Market (Public marketplace & advertiser)

| ID  | Module                                      | One-line purpose                                              | Release |
|-----|---------------------------------------------|---------------------------------------------------------------|---------|
| M25 | Public Discovery: Map & Search              | Geospatial search with clustering and rich filters            | V2.0    |
| M26 | Listing Detail & Availability Calendar      | Photographs, specifications, verified status, free-from dates | V2.0    |
| M27 | SEO & Content Engine                        | Programmatic location pages, rate guides, editorial content   | V2.0    |
| M28 | Advertiser Account, Shortlist & Enquiry     | Registration, saved boards, multi-board enquiry to owners     | V2.0    |
| M29 | Hold & Reservation                          | Time-boxed soft lock on inventory during negotiation          | V2.0    |
| M30 | Advertiser Campaign Portal & Proof Delivery | Live campaign status, photographic proof, documents, invoices | V3.0    |
| M31 | Audience & Impressions Data                 | Traffic, footfall and impression estimates per site           | V5.0    |

### Surface C — H360 Field, platform & extensibility

| ID  | Module                                     | One-line purpose                                                  | Release |
|-----|--------------------------------------------|-------------------------------------------------------------------|---------|
| M32 | H360 Field (mobile / PWA)                  | Offline-tolerant field capture: QR scan, photo, GPS, close job    | V1.1    |
| M33 | Super Admin & Platform Operations          | Tenant verification, moderation, subscription and platform health | V1.0    |
| M34 | Public API, Webhooks & Programmatic Supply | Partner integration and supply-side availability feeds            | V5.0    |
| M35 | White-label & Enterprise Deployment        | Custom domains, branding, SSO, dedicated environments             | V5.0    |

## 7.2 Scope at a glance by release

| Release | Modules delivered (new or materially extended)                                   | Count |
|---------|----------------------------------------------------------------------------------|-------|
| V1.0    | M01, M02, M03, M04, M05, M06, M07, M08, M09, M10, M33 (basic)                    | 11    |
| V1.1    | M11, M12, M32                                                                    | 3     |
| V2.0    | M25, M26, M27, M28, M29, M33 (verification & moderation)                         | 6     |
| V2.1    | M13, M14, M15                                                                    | 3     |
| V3.0    | M16, M17, M30, M33 (billing & payouts)                                           | 4     |
| V4.0    | M18, M19, M20, M21, M22, M04 (compliance matrix & scoring), M11 (SLA & dispatch) | 7     |
| V5.0    | M23, M24, M31, M34, M35                                                          | 5     |



## 8. Architectural Guardrails — Sprint 0 Non-Negotiables

### Read this before writing a single migration

The features in V4.0 and V5.0 will not be built for a year or more. Their **database relationships and service boundaries must exist from day one**. If these ten rules are followed, every future module is additive. If they are ignored, the system will require a destructive rewrite within 18 months — at exactly the moment the business is trying to scale. **These rules are not negotiable for schedule reasons.**

### 8.1 The ten rules

#### Rule 1 — Multi-tenancy from the first table

- Every business table carries a non-nullable `tenant_id`. No exceptions, including lookup and configuration tables that owners can customise.
- Enforce isolation at the data layer (PostgreSQL Row-Level Security policies), not only in application code. Application-layer-only isolation fails the first time someone writes a raw query or a report job.
- A single shared database with RLS is the default. Reserve the option of a dedicated schema or database per Tier A tenant for V5.0 (M35) by never hardcoding a connection string.
- Platform staff access to tenant data is itself an audited event (`TENANT_IMPERSONATION_STARTED`).

#### Rule 2 — The board is the root entity

- `boards` is the anchor of the domain. Campaigns, agreements, permits, incidents, work orders, documents, photos, leases and costs all resolve to one or more boards.
- Board identity is permanent. A board is never deleted — it is `retired` or `dismantled` with a reason and a date. Historical revenue and compliance records must remain queryable for at least seven years.
- Board **face** must be modelled from the start: a single structure can carry multiple advertising faces (e.g. a unipole with two faces, a gantry with three panels), each independently sold, priced and occupied. **Modelling one face per structure is the most common and most expensive early mistake in this domain.**

#### Rule 3 — Polymorphic document storage

- One `documents` table with `entity_type` + `entity_id`, serving boards, campaigns, agreements, permits, incidents, work orders, leases, vendors and clients.
- Every document carries: type, issuing authority (nullable), issue date, expiry date (nullable), version number, supersedes-document-id, signature status, uploader, checksum and storage key.
- Files live in object storage; the database stores metadata and a key. Never store binaries in PostgreSQL.
- Deletion is soft. A document referenced by a legal agreement can never be hard-deleted by a tenant user.

#### Rule 4 — Compliance is not a date field

- A board has **many** compliance records, not one licence expiry. Each has its own `governing_body`, `clearance_type`, `renewal_cycle_months`, `issue_date`, `expiry_date` and `status`.
- Clearance types are tenant-configurable (municipal advertising licence, traffic police NOC, structural stability certificate, electrical safety clearance, landowner permission letter, highway authority permission, fire NOC, and any others a specific city demands).
- Compliance status is computed, never manually set: `valid`, `expiring_soon`, `expired`, `missing`, `under_renewal`.

#### Rule 5 — Event bus from day one

- Modules never call each other directly. Every meaningful state change publishes an event to an asynchronous bus (Redis Streams or BullMQ at V1.0; assess a managed broker at V4.0 scale).

- Events are immutable, versioned (``schema_version``), and carry ``tenant_id``, ``actor_id``, ``entity_type``, ``entity_id``, ``occurred_at`` and a typed payload.
- Consumers are idempotent and independently retryable, with a dead-letter queue and an operator replay tool.
- The full event catalogue is in Section 10.1. **Adding a notification, a webhook or an analytics metric must never require touching the module that produced the event.**

### Rule 6 — Explicit state machines

- Board status, agreement status, campaign status, booking status, incident status, work order status, permit status and payment status are each defined as an explicit set of states plus allowed transitions, stored as configuration.
- Transitions are performed through a single service that validates legality, records the actor and reason, writes an audit entry and emits an event.
- Status strings must never be compared as free text in frontend code. Section 10.2 defines every machine.

### Rule 7 — Everything is auditable, nothing is destroyed

- Append-only ``activity_events`` table backs the universal Activity Feed and the per-entity timeline.
- Field-level change history is captured for legally relevant entities: agreements, permits, pricing, board specifications, ownership and campaign dates.
- Every audit entry records who, what, when, from-value, to-value, and (where the UI can supply it) why.

### Rule 8 — Business rules are configuration, not code

- Alert thresholds (90/60/30/15/7 days), incident categories, board types, illumination types, document types, clearance types, SLA targets and commission percentages are tenant-level configuration with sensible platform defaults.
- Any hardcoded enum that a customer might reasonably want to change is a defect.

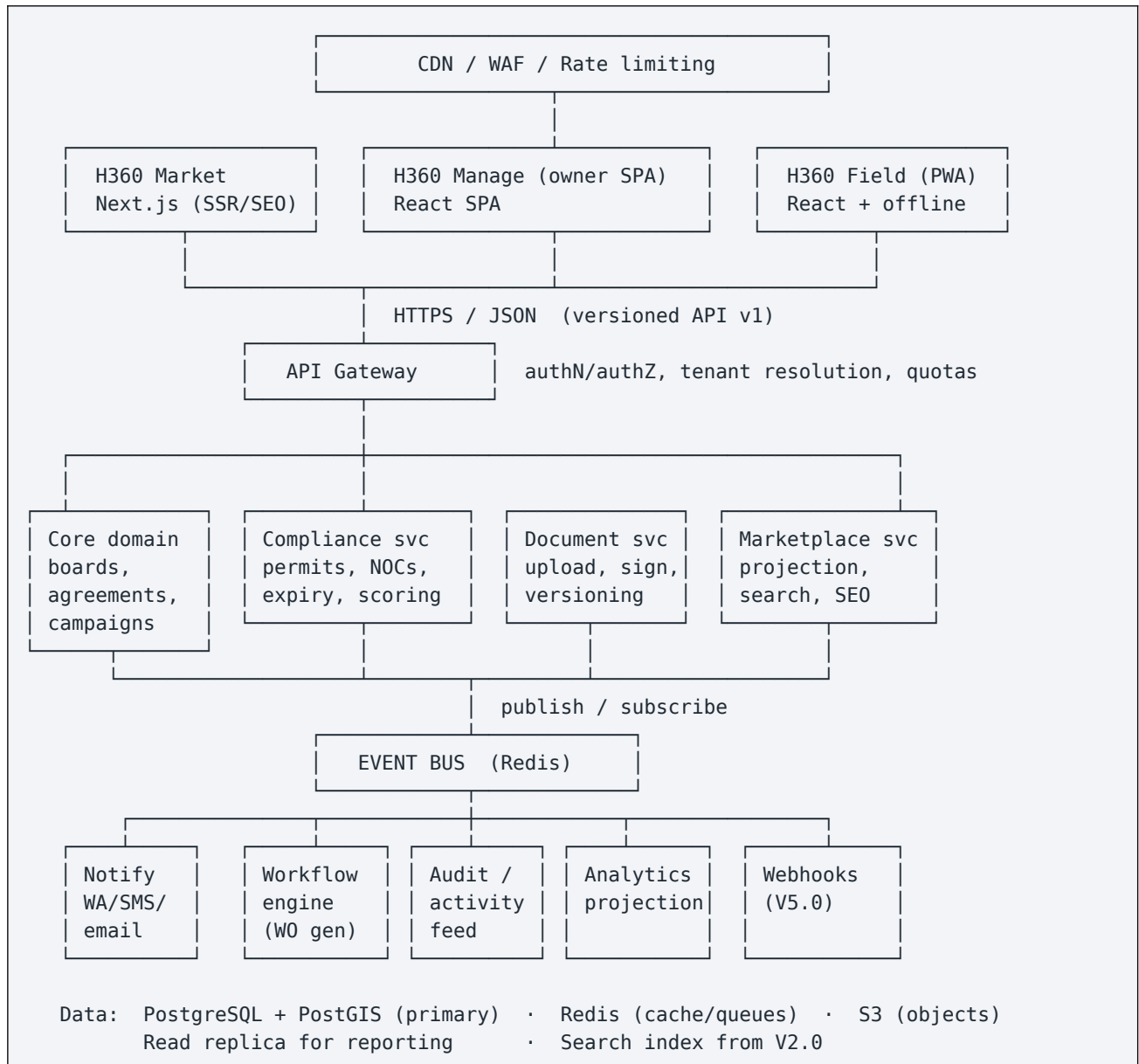
### Rule 9 — Attachments, comments and timeline on every entity

- Build three shared services once — attachments, comments/notes, and timeline — and compose them into every entity screen. Do not reimplement per module.
- This is what makes the product feel like an operating system rather than a set of forms.

### Rule 10 — API-first, with the marketplace as a consumer

- The owner platform, the marketplace and the field app are all clients of the same versioned REST/JSON API. No private back doors.
- Marketplace read traffic is served from a **projection** (a read-optimised, cached view of publishable inventory), not from live operational tables. Public traffic must never be able to degrade an owner's dashboard.
- Version the API from v1 and never break a published contract; the V5.0 partner API (M34) is then an authorisation change, not a rebuild.

## 8.2 Reference architecture



## 9. Domain Model & Core Entities

### 9.1 Entity register

The following entities must be **created or reserved** according to the "Schema" column. "Reserve" means the table and its foreign keys are designed and, where cheap, created empty — so that the later module is an additive change, not a migration of live data.

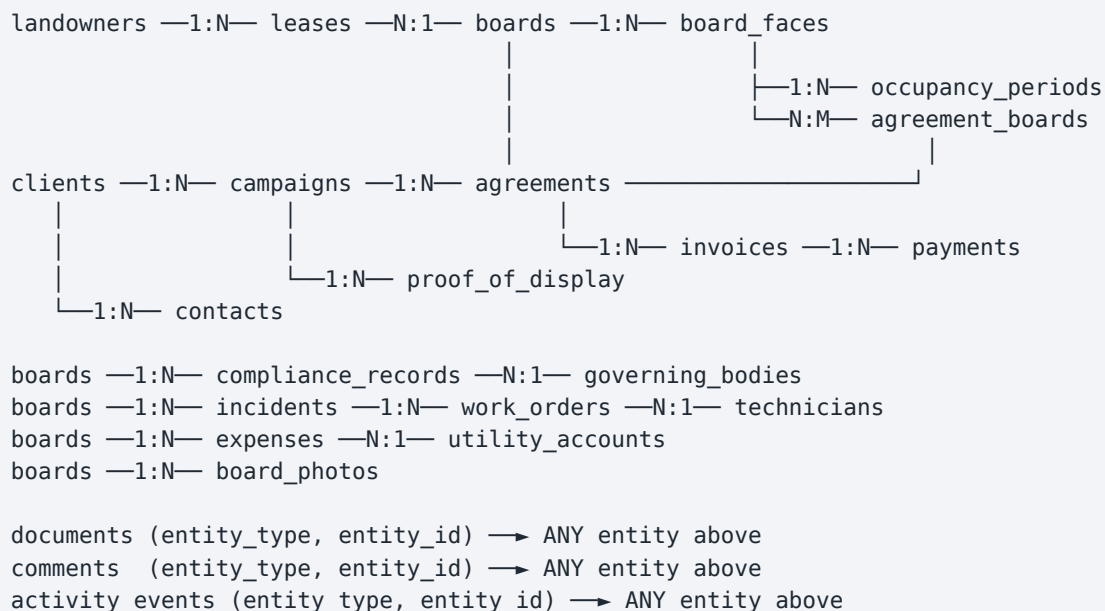
| Entity                                | Purpose                                                     | Schema                | UI   |
|---------------------------------------|-------------------------------------------------------------|-----------------------|------|
| tenants                               | A media owner company; the isolation boundary               | V1.0                  | V1.0 |
| users, roles, permissions, user_roles | Identity and RBAC                                           | V1.0                  | V1.0 |
| boards                                | The structure: location, type, ownership, lifecycle status  | V1.0                  | V1.0 |
| board_faces                           | Individually sellable advertising faces on a structure      | V1.0                  | V1.0 |
| board_specifications                  | Versioned dimensions, illumination, orientation, visibility | V1.0                  | V1.0 |
| board_photos                          | Day/night/approach imagery with capture metadata            | V1.0                  | V1.0 |
| clients                               | Advertiser companies the owner contracts with               | V1.0                  | V1.0 |
| contacts                              | People at clients, landowners, vendors and authorities      | V1.0                  | V1.0 |
| agreements                            | Commercial contracts; versioned, valued, dated              | V1.0                  | V1.0 |
| agreement_boards                      | Which faces an agreement covers, at what rate               | V1.0                  | V1.0 |
| compliance_records                    | One row per clearance per board, each with its own expiry   | V1.0                  | V1.0 |
| documents                             | Polymorphic file metadata for every entity                  | V1.0                  | V1.0 |
| occupancy_periods                     | Derived, queryable occupancy timeline per face              | V1.0                  | V1.0 |
| activity_events                       | Append-only universal activity feed                         | V1.0                  | V1.0 |
| notifications, notification_rules     | Outbound messages and their triggers                        | V1.0                  | V1.0 |
| comments                              | Threaded notes attachable to any entity                     | V1.0                  | V1.0 |
| import_jobs, import_rows              | Bulk onboarding with per-row validation results             | V1.0                  | V1.0 |
| incidents, incident_types             | Configurable incident capture and lifecycle                 | V1.1                  | V1.1 |
| proof_of_display                      | Installation evidence with GPS, timestamp, device           | V1.1                  | V1.1 |
| campaigns                             | A client's booking across many boards — the commercial unit | <b>V1.0 (reserve)</b> | V2.1 |
| leads, enquiries                      | Inbound marketplace demand                                  | V2.0                  | V2.0 |
| advertisers, advertiser_users         | Demand-side accounts (platform-level, not tenant-scoped)    | V2.0                  | V2.0 |
| holds                                 | Time-boxed soft locks on a face                             | V2.0                  | V2.0 |
| media_plans, media_plan_items         | Saved board sets before they become campaigns               | V2.1                  | V2.1 |
| quotations, quotation_items           | Priced proposals with validity windows                      | V3.0                  | V3.0 |

| Entity                                     | Purpose                                              | Schema                | UI   |
|--------------------------------------------|------------------------------------------------------|-----------------------|------|
| bookings                                   | Confirmed commercial commitment                      | V3.0                  | V3.0 |
| invoices, invoice_lines, payments          | Billing and collection with GST fields               | V3.0                  | V3.0 |
| work_orders, tasks                         | Field jobs generated from incidents and campaigns    | <b>V1.1 (reserve)</b> | V4.0 |
| technicians, crews                         | Field workforce and assignment                       | <b>V1.1 (reserve)</b> | V4.0 |
| landowners, leases, lease_payments         | Land rent obligations and payout history             | <b>V1.0 (reserve)</b> | V4.0 |
| expenses, utility_accounts, meter_readings | Electricity and operating costs per structure        | <b>V1.0 (reserve)</b> | V4.0 |
| vendors, purchase_orders, vendor_ratings   | Printers, fabricators, electricians, crane operators | <b>V1.1 (reserve)</b> | V4.0 |
| board_health_scores                        | Computed composite asset health snapshots            | V4.0                  | V4.0 |
| audience_metrics                           | Traffic, footfall and impression estimates           | Reserve               | V5.0 |
| api_clients, webhooks, webhook_deliveries  | Partner integration                                  | Reserve               | V5.0 |

### The four "reserve now, build later" decisions that matter most

**campaigns** (a client books 200 boards, not one — agreements must attach to a campaign, not directly to a board), **leases** (without landowner cost, profitability is fiction), **work\_orders** (incidents are useless without dispatch), and **board\_faces** (a two-sided unipole is two sellable units). Getting these four relationships wrong is what forces the 18-month rewrite.

## 9.2 Core relationships



## 9.3 Design notes the engineering team must honour

- Geospatial from the start.** `boards.location` is a PostGIS `geography(Point, 4326)` with a GiST index. Radius, bounding-box and corridor queries are core to both the owner map and the marketplace; a latitude/longitude float pair will not survive.



12. **Money as integers.** Store all monetary values in paise (``bigint``) with an explicit currency column. Never use floating point for rates, rents or commissions.
13. **Dates with timezone.** All timestamps are ``timestampz`` stored in UTC. Business dates (campaign start, permit expiry) are ``date`` in IST semantics; document the distinction and test around it.
14. **Occupancy is derived and materialised.** Compute occupancy periods from agreement dates via a scheduled job and store them, so calendars and availability searches are fast and consistent. Never compute availability ad hoc in two different places.
15. **Soft delete everywhere,** with ``deleted_at``, ``deleted_by`` and ``deletion_reason``. Reporting queries exclude soft-deleted rows by default via views.
16. **Externally-visible identifiers.** Use UUIDs as primary keys, plus a human-readable per-tenant code for boards (e.g. ``BLR-HSR-0142``) that field staff and clients can quote over the phone.
17. **Tenant-configurable taxonomies** live in tables with a ``tenant_id`` nullable — null means "platform default", a value means "tenant override".

## 10. Event Catalogue & State Machines

### 10.1 Event catalogue

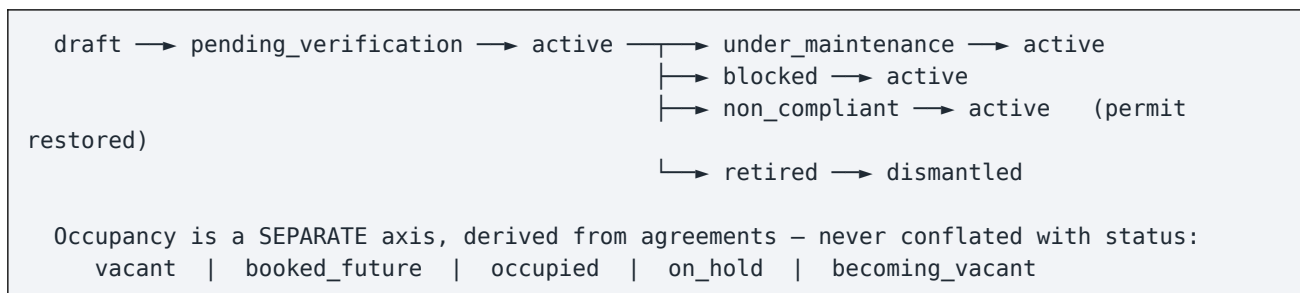
Every event below must exist in the bus by the release stated, even where no consumer uses it yet. Publishing early is nearly free; retro-fitting an event into a mature module is not.

| Event                                | Published when                                    | Initial consumers                                              | From |
|--------------------------------------|---------------------------------------------------|----------------------------------------------------------------|------|
| BOARD_CREATED                        | A structure is registered                         | Activity feed, analytics, marketplace projection               | V1.0 |
| BOARD_UPDATED                        | Specification, pricing or photos change           | Activity feed, marketplace projection, audit                   | V1.0 |
| BOARD_STATUS_CHANGED                 | Lifecycle transition (see 10.2)                   | Marketplace projection, dashboard, notifications               | V1.0 |
| AGREEMENT_CREATED / UPDATED          | A commercial contract is recorded or revised      | Occupancy engine, activity feed, audit                         | V1.0 |
| AGREEMENT_EXPIRING                   | Threshold days before end date                    | Notifications, CRM follow-up, marketplace pre-listing          | V1.0 |
| AGREEMENT_EXPIRED                    | End date passes                                   | Occupancy engine, marketplace publication, dashboard           | V1.0 |
| PERMIT_EXPIRING                      | Threshold days before a clearance expires         | Notifications, compliance dashboard                            | V1.0 |
| PERMIT_EXPIRED                       | A clearance lapses                                | <b>Marketplace suppression</b> , notifications, risk dashboard | V1.0 |
| PERMIT_RENEWED                       | A replacement clearance is uploaded and validated | Marketplace re-publication, compliance score                   | V1.0 |
| BOARD_BECOMING_VACANT                | Contract end enters the pre-listing window        | <b>Auto-listing to marketplace</b> , sales alert               | V1.0 |
| BOARD_VACANT                         | A face has no active agreement                    | Vacancy-loss reporting, sales alert                            | V1.0 |
| DOCUMENT_UPLOADED / SUPERSEDED       | Any document added or replaced                    | Activity feed, compliance engine, audit                        | V1.0 |
| USER_INVITED / ROLE_CHANGED          | Access changes                                    | Audit, security log                                            | V1.0 |
| TENANT_IMPERSONATION_STARTED / ENDED | Platform staff accesses tenant data               | Security audit (immutable)                                     | V1.0 |
| IMPORT_COMPLETED                     | A bulk import finishes                            | Notifications, onboarding progress tracker                     | V1.0 |
| INCIDENT_CREATED                     | An issue is reported on a structure               | Work order generation, notifications, health score             | V1.1 |
| INCIDENT_RESOLVED / CLOSED           | Repair verified                                   | Health score, SLA reporting, client notification               | V1.1 |
| PROOF_UPLOADED                       | Field evidence captured                           | Client notification, campaign status, advertiser portal        | V1.1 |
| ENQUIRY_RECEIVED                     | Advertiser submits an enquiry                     | Owner notification, CRM lead creation, SLA timer               | V2.0 |

| Event                                     | Published when                               | Initial consumers                             | From |
|-------------------------------------------|----------------------------------------------|-----------------------------------------------|------|
| HOLD_CREATED / EXPIRED / CONVERTED        | Soft lock lifecycle                          | Availability engine, owner notification       | V2.0 |
| LISTING_PUBLISHED / UNPUBLISHED           | Marketplace visibility changes               | Search index, SEO sitemap, analytics          | V2.0 |
| PROPOSAL_SENT / VIEWED / RESPONDED        | Media plan shared with a client and acted on | CRM, sales alerts, conversion analytics       | V2.1 |
| CAMPAIGN_CREATED / STARTED / ENDED        | Campaign lifecycle                           | Work orders, proof requirements, invoicing    | V2.1 |
| QUOTATION_ISSUED / ACCEPTED / REJECTED    | Quote lifecycle                              | Booking creation, CRM, analytics              | V3.0 |
| BOOKING_CONFIRMED / CANCELLED             | Commercial commitment                        | Occupancy, invoicing, commission, work orders | V3.0 |
| INVOICE_ISSUED / PAID / OVERDUE           | Billing lifecycle                            | Notifications, ageing report, payouts         | V3.0 |
| PAYMENT_RECEIVED / FAILED / REFUNDED      | Money movement                               | Finance ledger, commission, advertiser portal | V3.0 |
| WORK_ORDER_CREATED / ASSIGNED / COMPLETED | Field job lifecycle                          | Technician app, SLA, vendor performance       | V4.0 |
| LEASE_PAYMENT_DUE / PAID                  | Landowner obligation                         | Finance alerts, profitability                 | V4.0 |
| HEALTH_SCORE_CHANGED                      | Composite asset score crosses a band         | Operations dashboard, prioritisation          | V4.0 |

## 10.2 State machines

### 10.2.1 Board face status (the most-read status in the product)



#### A very common modelling error

"Available", "Booked", "Under Maintenance" and "Licence Expired" are **not one enum**. Lifecycle status (is this structure usable?), compliance status (is it legal?) and occupancy status (is it sold?) are three independent dimensions. A board can be `active` + `non\_compliant` + `occupied` simultaneously — and that combination is precisely the situation the owner most needs to see. Collapsing them into a single field is the fastest way to make the compliance engine useless.

### 10.2.2 Other machines

| Entity            | States                                                                           | Notes                                                                               |
|-------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Compliance record | missing → applied → under_renewal → valid → expiring_soon → expired → superseded | Computed nightly and on document upload; `expired` triggers marketplace suppression |

| Entity     | States                                                                                | Notes                                                                                 |
|------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Agreement  | draft → pending_signature → active → expiring → expired → terminated / renewed        | `terminated` requires a reason and is audited; `renewed` links to successor agreement |
| Campaign   | planned → confirmed → in_installation → live → completed → cancelled                  | Drives work order and proof-of-display requirements                                   |
| Hold       | active → converted → expired → released                                               | Auto-expires; releases inventory back to availability                                 |
| Quotation  | draft → sent → viewed → negotiating → accepted → rejected → expired                   | Validity window enforced                                                              |
| Booking    | pending_payment → confirmed → active → completed → cancelled → refunded               | Cancellation policy configurable per tenant                                           |
| Incident   | reported → triaged → assigned → in_progress → resolved → verified → closed / rejected | Verification requires an after-photo; SLA clock runs from `triaged`                   |
| Work order | created → assigned → accepted → en_route → on_site → completed → verified → closed    | GPS check-in required at `on_site`                                                    |
| Invoice    | draft → issued → partially_paid → paid → overdue → cancelled → credited               | Overdue is computed, not set                                                          |

### 10.3 Workflow examples (event chains)

```

AGREEMENT_EXPIRING (60 days)
├─ notify: account manager (in-app + email)
├─ CRM: create renewal follow-up task
└─ dashboard: add to "expiring contracts" widget

AGREEMENT_EXPIRING (30 days) — the vacancy loop —
├─ marketplace: publish face as "Available from <end_date + 1>"
├─ notify: sales team (WhatsApp) "Board BLR-HSR-0142 free from 01 Oct"
└─ analytics: start vacancy-risk timer

PERMIT_EXPIRED
├─ marketplace: UNPUBLISH the face immediately (never sell illegal inventory)
├─ notify: compliance officer + company admin (WhatsApp + email, high priority)
├─ board: set compliance_status = expired, recompute health score
└─ audit: immutable record with document references

INCIDENT_CREATED (severity = high)
├─ work order: auto-generate with SLA target from configuration
├─ dispatch: assign to on-call technician for the geography
├─ notify: technician (WhatsApp) + operations manager (in-app)
└─ if board is occupied: notify account manager (client may need informing)

```

These chains are implemented as **event subscribers**, not as code inside the agreement or permit modules. That is what makes the WhatsApp channel in V1.1, the client portal in V3.0 and the partner webhook in V5.0 additive rather than invasive.

## 11. Module Specifications — H360 Manage (Owner Platform)

Each module below states its purpose, its functional requirements and the release in which it ships. Requirements marked **[Core]** are release-blocking; **[Should]** may be deferred within the same release; **[Later]** is explicitly out of scope for that release and named here only to prevent scope creep arguments during the sprint.

### M01 — Identity, Tenancy & Access Control · V1.0

**Purpose:** guarantee that one media owner can never see another's data, and that within a tenant, people see only what their job requires.

- **[Core]** Tenant registration with company profile, GST number, registered address, logo and branding colours (used later in proposals and invoices).
- **[Core]** Email + password authentication with bcrypt/argon2 hashing, email verification, password reset, and mandatory OTP for privileged role changes.
- **[Core]** JWT access tokens (short-lived) plus refresh tokens with rotation and server-side revocation. Token payload carries `tenant\_id` and role claims.
- **[Core]** Role-based access control with the role catalogue in Section 4.3; permissions expressed as `module.action` (e.g. `permits.edit`, `pricing.view\_cost`).
- **[Core]** Row-Level Security policies in PostgreSQL keyed on the session tenant.
- **[Core]** User invitation flow, deactivation (never deletion), and a per-tenant audit of access changes.
- **[Should]** Geography-scoped roles (a supervisor sees only their district's boards).
- **[Later]** SAML/OIDC SSO and SCIM provisioning — V5.0 (M35).

### M02 — Asset Registry & Board 360 · V1.0

**Purpose:** create a permanent digital twin of every structure. This is the most important module in the product; every other module reads from it.

#### Board record

- **[Core]** Identity: system UUID, human-readable tenant code, name, structure type (unipole, billboard/hoarding, gantry, LED/DOOH, bus shelter, kiosk, pole kiosk, wall wrap, mall/indoor — tenant-configurable).
- **[Core]** Location: GPS point (PostGIS), street address, landmark, city, district, state, PIN, ward/zone, road name and road category (highway, arterial, internal).
- **[Core]** Faces: one or more sellable faces per structure, each with its own facing direction, dimensions (width × height, in feet, with computed square footage), illumination (front-lit, back-lit, non-lit, LED), visibility distance, traffic direction faced, and its own base rate.
- **[Core]** Commercials: card rate per face per month, floor rate (visible only to roles with `pricing.view\_floor`), printing and mounting charges, and rate effective-from date with full history.
- **[Core]** Photographs: day, night and approach views; at least one image mandatory before a face can be published to the marketplace; capture date recorded and surfaced as "last verified".
- **[Core]** Lifecycle: installation date, structure owner, ownership type (owned / leased-in / managed for third party), retirement with reason.
- **[Should]** Street-view URL, drone imagery, structural drawings, meter number, and a free-text "how to reach" note for field staff.

#### The Board 360 screen

A single page that answers everything about one structure. This screen is the product demo. It must be built in V1.0 with tabs that are empty-but-present for modules that ship later, so owners can see where the product is going.

| Panel                   | Shows                                                                                 | Available from           |
|-------------------------|---------------------------------------------------------------------------------------|--------------------------|
| Header                  | Code, name, type, lifecycle status, compliance status, occupancy status, health score | V1.0 (health score V4.0) |
| Location & map          | Pin, address, ward, road, directions link                                             | V1.0                     |
| Faces & specifications  | Each face with dimensions, illumination, orientation, rate                            | V1.0                     |
| Gallery                 | Day/night/approach photographs with capture dates                                     | V1.0                     |
| Occupancy timeline      | Visual calendar: past, current and booked-future periods, with client names           | V1.0                     |
| Compliance              | Every clearance, authority, expiry, document, renewal status                          | V1.0                     |
| Agreements              | Current and historical contracts, values, documents                                   | V1.0                     |
| Documents               | All attached files, versioned                                                         | V1.0                     |
| Activity timeline       | Every event ever recorded against this board                                          | V1.0                     |
| Incidents & maintenance | Open and historical issues, resolution photos                                         | V1.1                     |
| Proof of display        | Installation evidence per campaign                                                    | V1.1                     |
| Marketplace             | Published state, views, enquiries received                                            | V2.0                     |
| Landowner & costs       | Lease terms, rent, electricity, payout status                                         | V4.0                     |
| Profitability           | Revenue less costs, occupancy rate, vacancy loss                                      | V4.0                     |

### M03 — Document Vault · V1.0

**Purpose:** one repository for every file in the business, attachable to any entity, versioned and never silently lost.

- **[Core]** Upload PDF, JPG, PNG, DOCX, XLSX up to 25 MB per file; virus scan on ingest; server-side encryption at rest.
- **[Core]** Polymorphic attachment (`entity\_type`, `entity\_id`) with document type taxonomy: agreement, municipal licence, traffic police NOC, structural certificate, electrical clearance, landowner permission, NOC (other), tax receipt, electricity bill, invoice, installation photo, correspondence, other.
- **[Core]** Versioning: a new upload of the same document type supersedes rather than overwrites; the chain is browsable.
- **[Core]** Metadata: issuing authority, reference number, issue date, expiry date, amount, signature status, uploader, upload timestamp, checksum.
- **[Core]** Search and filter by entity, type, authority, expiry window and free text on reference number.
- **[Core]** Signed, expiring download URLs; no public object storage buckets, ever.
- **[Should]** Drag-and-drop bulk upload with auto-classification by filename pattern.
- **[Later]** OCR extraction of expiry dates and reference numbers (V4.0); e-signature integration (V3.0, M16).

### M04 — Compliance & Permit Engine · V1.0 (matrix and scoring V4.0)

**Purpose:** ensure no structure is ever operating on a lapsed clearance without somebody having been warned four times. **This is the single strongest reason a media owner will pay for the product.**

- **[Core]** Multiple independent clearance records per board, each with clearance type, governing body, reference number, issue date, expiry date, renewal cycle, fee paid and attached document.
- **[Core]** Tenant-configurable clearance types with a platform default set (municipal advertising licence, traffic police NOC, structural stability certificate, electrical safety clearance, landowner permission, highway authority permission).

- **[Core]** Computed status per clearance and a rolled-up compliance status per board and per tenant.
- **[Core]** Escalating alerts at configurable thresholds (default 90 / 60 / 30 / 15 / 7 days, then daily after expiry) with role-based recipients.
- **[Core]** Renewal workflow: mark as under renewal, record application date and acknowledgement, upload the new certificate, auto-close the old record as superseded.
- **[Core]** Risk dashboard: expired, expiring within 30 days, missing documentation — with one-click drill-down to the affected boards.
- **[Core] Hard rule:** a face whose mandatory clearances are expired is automatically unpublished from the marketplace and cannot be included in a new quotation without an explicit, audited override by a Company Admin.
- **[V4.0]** Compliance matrix view (boards × clearance types as a colour-coded grid) and a tenant compliance score.
- **[Should]** Per-city renewal checklist templates capturing what each municipality actually requires.

## M05 — Client & Agreement Management · V1.0

**Purpose:** know exactly which board is committed to whom, for how long, at what value — and never be surprised by an ending contract.

- **[Core]** Client records: company name, GST number, billing address, industry, contacts, payment terms, credit notes.
- **[Core]** Agreement records: reference number, client, campaign (nullable in V1.0, mandatory from V2.1), start date, end date, total value, payment schedule, agreement document, status.
- **[Core]** Agreement line items: which faces, at what negotiated rate, for what period — a single agreement may cover many faces with different dates.
- **[Core]** Versioning: rate revisions, extensions and terminations create new versions with the reason recorded and the prior version retained.
- **[Core]** Renewal alerts at 90 / 60 / 30 / 15 / 7 days with assignment to the owning sales executive.
- **[Core]** Full-text and faceted search across agreements by client, board, city, value range and expiry window.
- **[Should]** Payment schedule tracking with expected-versus-received flags (full invoicing in V3.0, M17).

## M06 — Availability & Occupancy Engine · V1.0

**Purpose:** the mechanism that makes the whole business work. It converts contract dates into live occupancy, future availability and the automatic marketplace listing that removes the owner's "call this number" flex board forever.

- **[Core]** Nightly (and on-change) computation of `occupancy\_periods` per face from all active agreements, holds and blocks.
- **[Core]** Occupancy states per face at any date: vacant, on\_hold, booked\_future, occupied, becoming\_vacant, blocked.
- **[Core]** Availability query API: "which faces are free for a continuous period between date X and date Y, in this city, of this type?" — this is the query the entire marketplace depends on and it must be indexed accordingly.
- **[Core]** Pre-listing window: configurable per tenant (default 30 days, allowed range 0–90). When a contract enters the window, the face publishes to the marketplace as "Available from &lt;date>" **while the current campaign remains live and undisturbed**.
- **[Core]** Manual blocks with reason (maintenance, owner reserve, structural work) that suppress marketplace availability.
- **[Core]** Conflict prevention: overlapping agreements on the same face are rejected with a clear error identifying the conflicting agreement.

- **[Core]** Vacancy register: every face currently vacant, with days vacant and computed revenue foregone at card rate.
- **[Should]** "Availability heat calendar" across the whole portfolio for the next 180 days.

#### Acceptance criterion the Product Owner will personally test

**Given** a face with an agreement ending on 30 September and a tenant pre-listing window of 30 days, **when** the system clock reaches 31 August, **then** the face appears on the public marketplace labelled "Available from 01 October", the current campaign remains marked live, the assigned sales executive receives a WhatsApp/email alert, and the event `BOARD\_BECOMING\_VACANT` is present in the activity feed. **If this does not work, V1.0 does not ship.**

## M07 — Notification & Alerting · V1.0

- **[Core]** Event-driven notification service with a rules table: event type → conditions → recipient roles → channels → template.
- **[Core]** Channels at V1.0: in-app notification centre and transactional email. WhatsApp and SMS from V1.1.
- **[Core]** Per-user notification preferences and a digest option (immediate / daily / weekly) to prevent alert fatigue.
- **[Core]** Delivery log with status, provider message ID, retries and failures, visible to tenant admins.
- **[Core]** Templates are data, not code; localisable (English at V1.0; regional language support assessed at V2.0).
- **[V1.1]** WhatsApp Business API via an approved provider, with pre-approved message templates. **Note for engineering:** template approval takes real calendar time — start the process in Sprint 1, not the sprint it is needed.
- **[Should]** Escalation: if a critical alert is unacknowledged for N days, notify the next role up.

## M08 — Activity Feed & Audit Trail · V1.0

- **[Core]** Append-only activity events consumed from the bus, rendered as a per-entity timeline and a tenant-wide feed.
- **[Core]** Field-level change history for agreements, permits, pricing, board specifications and campaign dates: who, what, when, from, to, why.
- **[Core]** Filterable by entity type, user, date range and event type; exportable to CSV for audits and disputes.
- **[Core]** Immutable at the application layer — no API path may update or delete an activity event.
- **[Should]** Retention policy configuration with legal-hold flag for entities under dispute.

## M09 — Owner Dashboard & Operational Reporting · V1.0

**Purpose:** the screen Rajesh opens every morning. If it is not worth opening daily, the product is not sticky and everything downstream fails.

- **[Core]** Portfolio summary: total structures, total faces, occupied, vacant, occupancy percentage, month-on-month trend.
- **[Core]** Risk strip: permits expired, expiring in 30 days, missing documents, boards with no photographs, boards with no rate set.
- **[Core]** Commercial strip: contracts ending in 30/60/90 days with value at risk; contracted revenue for the current and next month.
- **[Core]** Vacancy loss: faces vacant, days vacant, cumulative revenue foregone at card rate, ranked worst-first.
- **[Core]** Map view of the portfolio, colour-coded by occupancy and compliance, with clustering.
- **[Core]** Standard exports (CSV/Excel): inventory register, occupancy report, compliance register, contract register, vacancy report.



- **[Should]** Saved views and per-role default dashboards.

## M10 — Bulk Import & Data Onboarding · V1.0

**Purpose:** remove the single biggest adoption blocker. An owner with 300 boards will not type them in. **This module is a growth feature, not a utility.**

- **[Core]** Downloadable Excel/CSV template (Appendix D) covering boards, faces, specifications, rates, clients, agreements and permits.
- **[Core]** Guided import: upload → column mapping → validation preview with per-row errors and warnings → confirm → import → summary report.
- **[Core]** Validation rules: mandatory fields, coordinate sanity check (inside India, plausible for the stated city), date logic, duplicate detection by proximity (within 25 m) and by name similarity.
- **[Core]** Partial import: valid rows commit, invalid rows are returned as a downloadable correction file.
- **[Core]** Bulk photo upload with filename-to-board-code matching.
- **[Core]** Onboarding progress tracker: percentage of boards with coordinates, photographs, rates, permits and agreements — gamified, because data completeness is what makes the marketplace credible.
- **[Should]** Field-capture mode in the mobile app for surveying a board from site in under 90 seconds (V1.1, M32).

## M11 — Incident Management · V1.1 (dispatch & SLA V4.0)

**Purpose:** turn "the light on board 142 is fused and the client is angry" from a WhatsApp message into a tracked, measurable, closable record.

- **[Core]** Report an incident against a board or face from web or mobile, with photographs mandatory.
- **[Core]** **Tenant-configurable** incident categories, with platform defaults grouped as: Structural (pole damage, rust, foundation), Visibility (tree obstruction, encroachment, unauthorised posters), Electrical (lighting failure, wiring, LED malfunction, meter/power), Operational (wrong creative, torn flex, incorrect dimensions, late installation), Other.
- **[Core]** Severity (configurable, default low/medium/high/critical), reporter, description, affected campaign auto-linked if the face is occupied.
- **[Core]** Lifecycle per Section 10.2 with mandatory after-photograph before an incident can be verified and closed.
- **[Core]** Incident dashboard: open by severity, ageing, repeat offenders by board, resolution time distribution.
- **[Core]** If the affected face is occupied, the account manager is notified so the client hears it from us first.
- **[V4.0]** SLA targets per category, automatic technician dispatch, escalation on breach, technician performance analytics.
- **[Should]** Public incident reporting link (QR on the structure) so a client or member of the public can report a fault without an account.

## M12 — Proof of Display · V1.1

**Purpose:** answer "is my ad up?" before the client asks. This is the feature advertisers value most and the one that most reduces owner–client friction.

- **[Core]** Field capture of installation evidence: photographs taken **in-app** (camera only, gallery upload disabled by default) with GPS coordinates, device timestamp, server receipt timestamp and technician identity embedded as metadata.
- **[Core]** Geo-validation: capture location must be within a configurable radius (default 150 m) of the registered board location; outside that, the capture is flagged for supervisor review rather than rejected.
- **[Core]** Required evidence set per campaign, configurable: day shot, night shot (for lit sites), wide/approach shot, close-up of creative.

- **[Core]** Proof pack generation: a branded PDF per campaign with photographs, coordinates, timestamps, board specifications and a map, downloadable and shareable by link.
- **[Core]** Monitoring captures: periodic re-verification during a long campaign (configurable, e.g. fortnightly) to prove continued display.
- **[V3.0]** Automatic delivery into the advertiser's campaign portal (M30) the moment proof is uploaded.
- **[Should]** Image quality checks (blur, darkness, obstruction) with a re-capture prompt.

## M13 — Campaign Management · V2.1 (entity reserved V1.0)

**Purpose:** correct the most common structural error in OOH software. **A client does not book a board — a client runs a campaign across many boards.** Agreements, invoices, installations and proof all belong to the campaign.

- **[Core]** Campaign record: client, name, brand, objective, start and end dates, total value, status, owning sales executive.
- **[Core]** Many-to-many board/face membership with per-face dates and rates (faces may join and leave a campaign mid-flight).
- **[Core]** Campaign calendar and Gantt view across all member faces.
- **[Core]** Automatic generation of installation and removal tasks from campaign dates (work orders proper in V4.0, M18).
- **[Core]** Campaign-level proof pack aggregating evidence from every face.
- **[Core]** Extension, reduction and cancellation flows, each versioned and audited, with automatic occupancy recalculation.
- **[Should]** Campaign templates for repeat clients; multi-city campaign grouping.

## M14 — CRM & Sales Pipeline · V2.1

- **[Core]** Lead capture from marketplace enquiries (M28), manual entry and CSV import, with source attribution.
- **[Core]** Pipeline stages (configurable): new → qualified → proposal sent → negotiation → won / lost, with loss reasons.
- **[Core]** Client 360: contacts, campaign history, total spend, average deal size, agreements, outstanding invoices, communication log.
- **[Core]** Communication log: calls, meetings, emails and WhatsApp conversations recorded against the client, manually or via integration.
- **[Core]** Automated renewal follow-ups triggered by `AGREEMENT\_EXPIRING`, assigned to the account owner with a due date.
- **[Core]** Enquiry response SLA tracking — because a marketplace enquiry ignored for three days destroys advertiser trust in the whole platform.
- **[Should]** Sales targets and per-executive performance dashboard.

## M15 — Media Plan & Proposal Builder · V2.1

### This is the product's likely "killer feature" for the sales persona

Today a sales executive spends 3–4 hours building a PowerPoint for every brief: downloading photographs, copying dimensions, pasting map screenshots, retyping rates. Compressing that to under 10 minutes, with better output, is the most viscerally persuasive thing we can demonstrate in a sales meeting. **Prioritise it immediately after the marketplace launch.**

- **[Core]** Select faces from the map or list (with availability filtering) into a named media plan.
- **[Core]** Plan editing: add/remove faces, set period, apply discounts per face or per plan, add printing and mounting charges, compute totals with and without GST.

- **[Core]** Branded PDF export using the tenant's logo and colours: cover page, summary, map with all pins, one page per site (day and night photographs, specifications, location, rate), commercial summary and terms.
- **[Core]** PPTX export with the same content, editable by the sales team.
- **[Core]** Shareable interactive link: the client views the plan in a browser, sees each site on a map, and can approve or reject individual sites and add comments. Owner is notified of every view and action (`PROPOSAL\_VIEWED`).
- **[Core]** Conversion: an approved plan becomes a campaign (M13) and, from V3.0, a quotation (M16) in one click, with no re-keying.
- **[Should]** Plan templates by budget or city; automatic alternative suggestions when a selected face is unavailable for the requested period.

## M16 — Quotation, Booking & Contract Lifecycle · V3.0

- **[Core]** Quotation generation from a media plan with validity period, terms, taxes and approval workflow above a configurable value threshold.
- **[Core]** Negotiation history: versions, revised rates, reasons, all audited.
- **[Core]** Acceptance converts to a booking, which places holds on the relevant faces and blocks conflicting sales immediately.
- **[Core]** Agreement generation from tenant-configurable templates with merge fields (client, faces, dates, rates, terms), output as PDF.
- **[Core]** E-signature integration (Indian providers supporting Aadhaar eSign, with an international fallback) including signature status tracking and a completion certificate stored in the vault.
- **[Core]** Distribution: email, WhatsApp and download from within the platform.
- **[Core]** Cancellation and amendment flows with configurable policy and automatic occupancy release.
- **[Should]** Separate template sets for client sales agreements and landowner lease agreements.

## M17 — Invoicing, Payments & GST · V3.0

- **[Core]** Invoice generation from bookings and payment schedules, with GST treatment (CGST/SGST/IGST determined by place of supply), HSN/SAC codes, invoice numbering series per tenant and financial year.
- **[Core]** Payment collection via a gateway supporting UPI, cards, net banking and NEFT/RTGS reconciliation.
- **[Core]** Receipts, credit notes, partial payments, advance/deposit handling and refunds.
- **[Core]** Accounts receivable ageing, overdue reminders on a configurable schedule, and a collections worklist.
- **[Core]** Platform commission calculation on marketplace-originated bookings, with a clear, auditable statement to the owner showing exactly what was charged and why.
- **[Core]** Export to Tally and Zoho Books formats. **We do not replace the accountant's system; we feed it.**
- **[Should]** E-invoicing (IRN/QR) support for tenants above the statutory turnover threshold. **Confirm current thresholds with a tax advisor before build — they change.**
- **[Later]** Escrow / payment protection for advertisers — assess at V4.0 with legal advice on money-transmission implications.

## M18 — Work Order & Field Force Management · V4.0

- **[Core]** Work orders generated automatically from incidents (M11) and campaign dates (M13), or created manually.
- **[Core]** Types: install creative, replace creative, remove creative, repair lighting, structural repair, clean structure, routine inspection, tree pruning follow-up, permit-related site work.
- **[Core]** Assignment to internal technicians, crews or external vendors (M20), with skill and geography matching.

- **[Core]** Mobile execution: accept, navigate, GPS check-in on site, before photographs, materials used, after photographs, complete.
- **[Core]** SLA targets per work order type with breach alerts and escalation.
- **[Core]** Scheduling board: today's jobs by technician and geography, with drag-and-drop reassignment.
- **[Core]** Performance analytics: jobs completed, average time to close, first-time-fix rate, SLA compliance, per technician and per vendor.
- **[Should]** Route optimisation for multi-stop days; material/inventory consumption tracking (flex rolls, lamps).

## M19 — Landowner Lease & Expense Ledger · V4.0 (entities reserved V1.0)

**Purpose:** without this module, every profitability number in the product is fiction. A hoarding sits on somebody else's land and consumes somebody's electricity.

- **[Core]** Landowner records: individual, company, government body or authority; contacts; bank and payment details; PAN/GST where applicable.
- **[Core]** Lease records per board: start and end dates, tenure, rent basis (fixed monthly, fixed annual, percentage revenue share, hybrid), amount, escalation clause and percentage, payment frequency, notice period, renewal terms, lease document.
- **[Core]** Payment schedule generation, due alerts (**a missed landowner payment can mean eviction and structure removal — treat as critical severity**), payment recording and receipts.
- **[Core]** Utility accounts per structure: connection/consumer number, meter number, provider, sanctioned load, tariff category; monthly bill capture and payment tracking.
- **[Core]** Other operating expenses per board: maintenance, repainting, security, municipal fees, insurance.
- **[Core]** Cost roll-up feeding M22 profitability.
- **[Should]** Landowner portal (read-only) showing lease terms and payout history — a retention lever for the owner's own landlord relationships.

## M20 — Vendor & Procurement · V4.0

- **[Core]** Vendor master: printers, flex fabricators, electricians, installers, crane and hydraulic operators, security, structural engineers; with categories, rate cards, GST details and service geography.
- **[Core]** Purchase orders linked to work orders and campaigns, with approval thresholds.
- **[Core]** Vendor invoice capture, matching against POs and work order completion, and payment status.
- **[Core]** Performance scoring: cost variance, on-time delivery, rework rate, quality rating.
- **[Should]** Vendor portal for accepting jobs and uploading completion evidence.

## M21 — Board Health Score · V4.0

**Purpose:** give an operations manager a single number that ranks 1,000 structures by how much they need attention today.

- **[Core]** Composite score (0–100) from weighted, **configurable** components: compliance validity, structural condition (open structural incidents), illumination status, visibility (obstruction incidents), occupancy performance, contract status, documentation completeness, days since last physical verification.
- **[Core]** Bands (healthy / attention / critical) with a colour language used consistently across dashboards and maps.
- **[Core]** Score history and trend per board; `HEALTH\_SCORE\_CHANGED` event on band transition.
- **[Core]** Portfolio view sorted by score, and a "worst 20 boards this week" operations worklist.

## M22 — Financial Intelligence & Profitability · V4.0

- **[Core]** Revenue recognition per face per month from agreements and bookings.

- **[Core]** Cost allocation per board: land rent, electricity, maintenance, municipal fees, installation, insurance, depreciation of structure.
- **[Core]** Net margin per board, per face, per district, per city, per client and per structure type.
- **[Core]** Vacancy loss report: days vacant × card rate, with rolling 12-month totals and worst-performer ranking.
- **[Core]** Client profitability and concentration analysis (revenue share of top 5 clients — a genuine business risk metric).
- **[Core]** Forecast: contracted revenue for the next 12 months plus expected renewals, versus fixed cost base.
- **[Should]** Break-even analysis per structure; payback tracking on newly built structures.

## M23 — Business Intelligence & Heatmaps · V5.0

- Geographic heatmaps: revenue density, occupancy, demand (marketplace searches and enquiries), vacancy concentration.
- Temporal analysis: seasonality by month, category and city; renewal probability by client segment.
- Benchmark reports: anonymised, aggregated market rates and occupancy by city and format. **Requires an explicit, opt-in data policy — see Section 23, OQ-6.**
- Custom report builder with scheduled email delivery; export to Excel and PDF.

## M24 — AI Pricing & Vacancy Prediction · V5.0

- Rate recommendation per face from location, format, historical realised rates, occupancy, seasonality and local demand signals — presented as a **suggestion with an explanation**, never as an automatic price change.
- Vacancy prediction: probability that a given face will be unsold in the next 30/60/90 days, driving proactive sales worklists.
- Renewal propensity per client; churn-risk alerts for the account manager.
- **Guardrail:** every model output must be explainable, overridable and logged. Owners will not accept an unexplained number attached to their most valuable asset.

## 12. Module Specifications — H360 Market (Marketplace & Advertiser)

### Launch gate

None of this ships until the V2.0 entry criteria in Section 16.6 are met. A marketplace launched with thin, stale inventory permanently damages the brand with the exact buyers we most need. **Inventory density and data freshness are launch blockers, not "nice to have".**

### M25 — Public Discovery: Map & Search · V2.0

- **[Core]** Full-screen interactive map with marker clustering, capable of rendering 25,000+ boards without degradation, served from the marketplace projection.
- **[Core]** Filters: city, locality/area, highway or road, structure type, face dimensions and size band, illumination, availability date range, monthly budget range, orientation, and "verified" status.
- **[Core]** Sorting: relevance, price ascending/descending, size, availability date, recently verified.
- **[Core]** Draw-on-map area selection and radius search around a point or landmark (PostGIS-backed).
- **[Core]** List and map views synchronised; shareable URLs that preserve the full filter state (essential for both SEO and agency workflows).
- **[Core]** Availability-aware results: entering a campaign period filters to faces genuinely free for that continuous period.
- **[Should]** Corridor search along a route (e.g. "all boards on the Bengaluru-Mysuru highway"); saved searches with email alerts on new matching inventory.

### M26 — Listing Detail & Availability Calendar · V2.0

- **[Core]** Photograph gallery (day, night, approach) with capture dates displayed.
- **[Core]** Specifications: dimensions, square footage, illumination, orientation, visibility distance, structure type, road and traffic context.
- **[Core]** Location: map pin, locality, landmark, and approximate address. **Exact coordinates are shown only to registered, verified advertisers** — an anti-scraping and owner-protection measure.
- **[Core]** Pricing display governed by a per-tenant policy: show card rate, show a rate band, or "price on request". Default is show card rate; **transparency is our differentiator and the sales team should actively push owners toward it.**
- **[Core]** Availability calendar for the next 180 days showing available, booked and on-hold periods, plus a prominent "Available from &lt;date&gt;" badge.
- **[Core]** Trust signals: "verified by HOARDINGS360", last verified date, permit-valid indicator, owner rating (from V3.0), and response-time indicator for the owner.
- **[Core]** Actions: add to shortlist, request quote, request hold, share, download a one-page site PDF.
- **[Should]** Similar or nearby alternatives, and "others also enquired about" suggestions.

### M27 — SEO & Content Engine · V2.0

**Purpose:** the cheapest durable demand channel available to us. Advertisers search "hoarding advertising cost in Kochi" long before they call an agency. **This module is jointly owned by Product and Marketing and must be specified with the marketing lead, not by engineering alone.**

- **[Core]** Server-side rendered, programmatically generated landing pages: city, locality, highway, structure type and city × type combinations (e.g. "Unipole advertising in Ernakulam").
- **[Core]** Each page carries real inventory counts, indicative price ranges, a map, sample sites and internal links — thin auto-generated pages will be penalised, so every template must include genuine, differentiated data.

- **[Core]** Structured data markup for local business and product/offer where legitimately applicable; XML sitemaps regenerated on listing changes; canonical URL discipline; robots rules that permit listing pages but block filter-permutation crawl traps.
- **[Core]** Core Web Vitals budget: LCP under 2.5 s on a mid-range Android device on 4G.
- **[Core]** Editorial content hub: rate guides by city, format explainers, permit and compliance guidance, campaign case studies.
- **[Should]** Multilingual pages for priority regional languages once English traffic is proven.

## M28 — Advertiser Account, Shortlist & Enquiry · V2.0

- **[Core]** Advertiser registration with email/mobile OTP verification; company profile; optional GST for later invoicing.
- **[Core]** Shortlist/wishlist across owners, with notes, campaign period and budget attached.
- **[Core]** Multi-board enquiry: one submission spanning several owners, automatically split into per-owner enquiries so no owner sees another owner's quoted price.
- **[Core]** Enquiry inbox for the advertiser with owner responses, quoted rates and threaded messaging.
- **[Core]** Owner response SLA display and enforcement: enquiries unanswered beyond the SLA are escalated to the owner's admin and, if chronic, affect marketplace ranking.
- **[Core]** Anti-spam and lead quality: OTP verification, rate limiting, duplicate detection, and platform moderation of suspicious accounts.
- **[Should]** Brief upload ("here is my requirement, find me boards") routed to a human planner in early releases — a high-conversion, low-technology growth lever.

## M29 — Hold & Reservation · V2.0

- **[Core]** Advertiser requests a hold on one or more faces; owner accepts or declines within a configurable window.
- **[Core]** Accepted holds soft-lock the face for a configurable duration (default 48 hours, owner-adjustable up to 7 days), suppressing it from other buyers' availability.
- **[Core]** Automatic expiry with notification to both sides; conversion to booking (V3.0) or manual release.
- **[Core]** Hold abuse controls: limits per advertiser, and a hold-to-conversion metric visible to owners.

## M30 — Advertiser Campaign Portal & Proof Delivery · V3.0

- **[Core]** Campaign dashboard: sites booked, period, status per site (pending installation, live, completed), days remaining.
- **[Core]** Automatic delivery of proof-of-display evidence (M12) as it is captured, with a downloadable campaign proof pack.
- **[Core]** Documents: agreement, invoices, receipts; payment status and history.
- **[Core]** Issue raising: the advertiser can report a problem with a site (unlit, damaged, wrong creative), which creates an incident in the owner's system with the client flagged as reporter.
- **[Should]** Post-campaign report with site-level evidence and a summary suitable for the advertiser's own client.

## M31 — Audience & Impressions Data · V5.0

- Traffic and footfall estimates per site from mapping providers, mobility data partners or municipal data where available.
- Impression and reach estimates per site and per plan, with a clearly published methodology and confidence level.

- **Guardrail:** never present a modelled number as measured. Publishing an unsubstantiated impressions figure would be the fastest way to lose credibility with professional agency buyers.



## 13. Module Specifications — Field, Platform & Cross-Cutting

### M32 — H360 Field (Mobile / PWA) · V1.1

**Build as an installable PWA first**, not native apps. Field staff use varied low-to-mid-range Android devices, storage is scarce, and app-store release cycles will slow us down during the period when we are iterating fastest. Reassess native at V4.0 if camera, background sync or push limitations prove blocking.

- **[Core]** Authentication with long-lived session suitable for field use; device binding for technicians.
- **[Core]** QR code on every structure (generated in M02, printed on a weatherproof sticker) → scan opens that board instantly.
- **[Core]** Today's tasks list, grouped by geography, with navigation hand-off to the device map app.
- **[Core]** In-app camera capture with automatic GPS, timestamp and technician stamping; image compression before upload.
- **[Core] Offline tolerance:** capture works without connectivity; captures queue locally and sync automatically when a connection returns, with clear pending/synced indicators. Field sites frequently have poor coverage — this is a core requirement, not an enhancement.
- **[Core]** Board survey mode: capture a new board from site (photos, GPS, dimensions, type) in under 90 seconds, for onboarding drives.
- **[Core]** Incident reporting with photographs from the field.
- **[V4.0]** Work order acceptance, check-in/check-out, materials used, completion with evidence.
- **[Should]** Low-data mode; regional language interface for technician roles.

### M33 — Super Admin & Platform Operations · V1.0, extended V2.0 and V3.0

| Capability             | Detail                                                                                                                                                  | Release             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Tenant management      | Create, suspend, configure tenants; plan assignment; feature flags per tenant                                                                           | V1.0                |
| Platform health        | Job queues, event bus lag, failed notifications, error rates, storage usage                                                                             | V1.0                |
| Support impersonation  | Time-boxed, reason-required, fully audited access to a tenant                                                                                           | V1.0                |
| Owner verification     | Review company registration, GST, sample permits and site ownership evidence before any listing goes public; issue the "Verified" badge                 | V2.0                |
| Listing moderation     | Review flagged listings, photograph quality, duplicate boards across tenants (the same structure claimed by two owners — expect this and design for it) | V2.0                |
| Content management     | SEO pages, editorial content, cities, taxonomy defaults                                                                                                 | V2.0                |
| Dispute management     | Case file combining enquiry, booking, proof and communications                                                                                          | V3.0                |
| Subscription & billing | Plans, invoices, dunning, upgrades, commission statements, payouts                                                                                      | V3.0                |
| Platform analytics     | Supply growth, data completeness, demand funnel, GMV, cohort retention                                                                                  | V2.0, extended V3.0 |

### M34 — Public API, Webhooks & Programmatic Supply · V5.0

- Versioned REST API with OAuth2 client credentials, per-client rate limits and usage analytics.
- Read endpoints for inventory, availability and pricing; write endpoints for enquiries and bookings.
- Outbound webhooks for the event catalogue, with signed payloads, retries and a delivery log.

- Supply-side feed compatible with programmatic DOOH partners and agency planning tools; assess adoption of prevailing OOH data interchange standards at design time.
- Developer documentation portal and sandbox environment.

## M35 — White-label & Enterprise Deployment · V5.0

- Custom domain, full brand theming, and tenant-branded advertiser-facing pages.
- SAML/OIDC SSO, SCIM provisioning, IP allow-listing, configurable data retention.
- Optional dedicated database or region for enterprise tenants; documented data-residency guarantees.
- Contractual SLA with credits, named support manager and quarterly business reviews.

### 13.3 RBAC permission matrix (V1.0 baseline)

R = read · W = create/edit · A = approve or administer · — = no access. Extend this table, do not replace it, as modules ship.

| Capability                 | Co. Admin | Ops Mgr | Sales | Compliance | Finance | Supervisor | Technician |
|----------------------------|-----------|---------|-------|------------|---------|------------|------------|
| Boards & specifications    | A         | W       | R     | R          | R       | R          | R          |
| Card rate (view)           | A         | R       | R     | —          | R       | —          | —          |
| Floor rate & cost data     | A         | R       | —     | —          | R       | —          | —          |
| Clients & agreements       | A         | R       | W     | R          | R       | —          | —          |
| Compliance records         | A         | R       | R     | W          | R       | R          | —          |
| Documents                  | A         | W       | R     | W          | R       | R          | —          |
| Incidents                  | A         | W       | R     | R          | —       | W          | W          |
| Work orders (V4.0)         | A         | A       | R     | —          | R       | W          | W (own)    |
| Proof of display           | A         | W       | R     | R          | —       | W          | W (own)    |
| Marketplace publication    | A         | W       | W     | R          | —       | —          | —          |
| Invoices & payments (V3.0) | A         | R       | R     | —          | W       | —          | —          |
| Landowner leases (V4.0)    | A         | R       | —     | R          | W       | —          | —          |
| Users & roles              | A         | —       | —     | —          | —       | —          | —          |
| Audit log                  | A         | R       | —     | R          | R       | —          | —          |

## 14. Non-Functional Requirements

### 14.1 Performance

| Area                      | Target                                                           | Measured as                |
|---------------------------|------------------------------------------------------------------|----------------------------|
| Owner dashboard load      | < 2.0 s at the 95th percentile with 5,000 boards in the tenant   | Real user monitoring       |
| Marketplace map render    | < 3.0 s to interactive with 25,000 boards in view (clustered)    | Lab + RUM                  |
| Availability search       | < 800 ms at the 95th percentile for a city-wide date-range query | API latency metric         |
| SEO landing pages         | LCP < 2.5 s on mid-range Android over 4G                         | Core Web Vitals field data |
| Photo upload (field, 4G)  | < 8 s per image including compression                            | Client instrumentation     |
| Bulk import               | 5,000 rows validated and committed in < 3 minutes                | Job duration metric        |
| Proof pack PDF generation | < 20 s for a 25-site campaign                                    | Job duration metric        |

### 14.2 Scale targets (design for, do not build for)

- 500 tenants · 150,000 boards · 300,000 faces · 5,000 concurrent owner-platform users.
- 50,000 monthly marketplace visitors at V2.0, designed to absorb 500,000 without re-architecture.
- 2 million activity events per month; 10 TB of object storage within 24 months (photographs dominate — enforce compression and lifecycle policies from day one).

### 14.3 Availability & resilience

- Target 99.5% monthly availability for the owner platform at V1.0, rising to 99.9% at V4.0 with a contractual SLA for Enterprise.
- The public marketplace is served from cache/CDN and a read projection; it must remain browsable (read-only) during a primary database incident.
- Automated daily backups with point-in-time recovery; **restore tested quarterly** — an untested backup is not a backup.
- RPO 15 minutes, RTO 4 hours at V1.0; RPO 5 minutes, RTO 1 hour at V4.0.
- Graceful degradation: if the notification provider or map provider is down, core workflows must continue and queue.

### 14.4 Security

- TLS 1.2+ everywhere; HSTS; secure and httpOnly cookies; strict CSP on all web surfaces.
- Encryption at rest for database and object storage; secrets in a managed secret store, never in the repository.
- Row-Level Security enforced tenant isolation, with an automated test suite that attempts cross-tenant access on every build. **This test suite is release-blocking.**
- Input validation and parameterised queries throughout; output encoding; file-type and content verification on upload; anti-virus scanning.
- Rate limiting and bot protection on all public endpoints; CAPTCHA on enquiry submission.
- OWASP Top 10 review before every major release; third-party penetration test before V2.0 (public exposure) and annually thereafter.

- Least-privilege infrastructure access; MFA mandatory for all platform staff; all production access logged.
- Documented incident response plan with breach notification obligations mapped.

## 14.5 Privacy, data protection & regulatory

- Comply with India's **Digital Personal Data Protection Act, 2023** and rules made under it: lawful purpose, consent notices in clear language, purpose limitation, data-principal rights (access, correction, erasure, grievance redressal), and breach notification. **Engage counsel before V2.0 launch; the rules and timelines have been evolving and must be confirmed at build time.**
- Publish a privacy notice, terms of use, a marketplace listing policy and a data-sharing policy before any public launch.
- Explicit, revocable, opt-in consent from tenants for any use of their data in aggregated benchmark products (M23).
- Data residency in India by default; document the hosting region in customer contracts.
- Retention policy: operational data retained for the contract term plus seven years for legal/tax defensibility; personal data of marketplace users deleted or anonymised on request, subject to statutory retention.
- GST and invoicing compliance validated by a tax advisor before V3.0.

## 14.6 Usability, accessibility & localisation

- Mobile-responsive across all surfaces; owner platform usable on a tablet; field app designed mobile-first for one-handed use in sunlight (high contrast, large touch targets).
- WCAG 2.1 AA as the target for public marketplace pages.
- English at V1.0. Assess Malayalam, Tamil, Kannada, Telugu and Hindi for the field app and marketplace at V2.0 — field technicians are the users most likely to need it.
- Indian formatting conventions throughout: ₹ with lakh/crore grouping, DD-MM-YYYY dates, Indian mobile number validation, IST display.
- Every destructive action requires confirmation; every long-running action shows progress; every error message states what to do next.

## 14.7 Observability & quality

- Centralised structured logging with correlation IDs across API, workers and event consumers.
- Metrics and alerting on queue depth, event lag, notification failures, job failures, API error rate and database saturation.
- Error tracking with release tagging; user-session replay on the marketplace (privacy-scrubbed).
- Automated tests: unit coverage above 70% on domain logic, integration tests on every API endpoint, end-to-end tests on the ten critical journeys in Section 18, and the tenant-isolation suite.
- CI/CD with automated migration checks; every migration must be reversible or accompanied by a documented forward-fix plan.

## 15. Technology Stack, Integrations & Environments

### 15.1 Recommended stack

| Layer                    | Recommendation                                                                                                | Rationale                                                                                                                                                  |
|--------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Marketplace frontend     | Next.js (React) with SSR/ISR                                                                                  | SEO is a primary demand channel; server rendering is non-negotiable for M27                                                                                |
| Owner platform frontend  | React SPA (Vite) + TypeScript                                                                                 | Rich, stateful, authenticated application; no SEO requirement                                                                                              |
| Field app                | React PWA with service worker and IndexedDB queue                                                             | Offline capture, no app-store release friction, one codebase                                                                                               |
| Backend API              | Node.js with NestJS (TypeScript)                                                                              | Strong module boundaries, DI, first-class queue and event support; one language across the stack simplifies a small team                                   |
| Alternative backend      | Python with FastAPI                                                                                           | Acceptable if the team is stronger in Python and later ML work (M24) is expected in-house. <b>Decide in Sprint 0 and do not revisit</b> (Section 23, OQ-3) |
| Primary database         | PostgreSQL 15+ with PostGIS                                                                                   | Geospatial querying is core; RLS provides tenant isolation; mature and inexpensive                                                                         |
| Cache, queues, event bus | Redis (Streams / BullMQ)                                                                                      | Sufficient to V4.0; reassess a managed broker at scale                                                                                                     |
| Search (V2.0)            | PostgreSQL full-text initially; OpenSearch/Meilisearch if needed                                              | Do not add a search cluster before the query patterns justify it                                                                                           |
| Object storage           | AWS S3 (or equivalent) with lifecycle policies and CDN                                                        | Photograph-heavy workload; cost control through tiering                                                                                                    |
| Maps                     | Google Maps Platform or Mapbox GL JS                                                                          | Evaluate on India data quality, clustering performance and cost at 25k+ markers (Section 23, OQ-4)                                                         |
| Messaging                | WhatsApp Business API via an approved BSP · transactional email provider · SMS provider with DLT registration | WhatsApp is the channel this industry actually reads. DLT template registration for SMS takes weeks — start early                                          |
| E-signature (V3.0)       | Indian eSign provider (Aadhaar eSign) with an international fallback                                          | Legal enforceability under Indian law                                                                                                                      |
| Payments (V3.0)          | Indian payment gateway supporting UPI, cards, net banking, and payouts                                        | Also required for landowner payouts and owner settlements                                                                                                  |
| Documents & exports      | Server-side PDF (headless browser rendering) and PPTX generation                                              | Required by M15 proposal exports and M12 proof packs                                                                                                       |
| Hosting                  | AWS ap-south-1 (Mumbai), containerised, infrastructure as code                                                | Data residency, latency, managed services                                                                                                                  |
| Analytics                | Product analytics tool + a warehouse from V3.0                                                                | Section 20 instrumentation depends on this being in place from V1.0                                                                                        |

### 15.2 Third-party integrations by release

| Integration   | Purpose                   | Release | Lead time risk                           |
|---------------|---------------------------|---------|------------------------------------------|
| Maps provider | Geocoding, map rendering, | V1.0    | Low — but negotiate pricing before scale |

| Integration                      | Purpose                                              | Release | Lead time risk                                                                           |
|----------------------------------|------------------------------------------------------|---------|------------------------------------------------------------------------------------------|
|                                  | directions                                           |         |                                                                                          |
| Email (transactional)            | Alerts, invitations, reports                         | V1.0    | Low — warm up the sending domain early                                                   |
| Object storage + CDN             | Photographs, documents, proof packs                  | V1.0    | Low                                                                                      |
| WhatsApp Business API (BSP)      | Expiry alerts, field dispatch, client proof delivery | V1.1    | <b>High — business verification and template approval take weeks. Start in Sprint 1.</b> |
| SMS provider                     | Fallback alerts, OTP                                 | V1.1    | <b>Medium — DLT entity and template registration required in India</b>                   |
| Search index                     | Marketplace search at scale                          | V2.0    | Low                                                                                      |
| E-signature provider             | Agreements and leases                                | V3.0    | Medium — commercial and legal review                                                     |
| Payment gateway                  | Collections, refunds, payouts                        | V3.0    | <b>Medium — KYC, merchant onboarding and settlement account setup</b>                    |
| Accounting export (Tally / Zoho) | Finance handoff                                      | V3.0    | Low                                                                                      |
| Mobility / traffic data partner  | Impressions and footfall                             | V5.0    | High — commercial negotiation                                                            |
| DOOH CMS / programmatic partners | Digital screen supply                                | V5.0    | High                                                                                     |

### 15.3 Environments and delivery

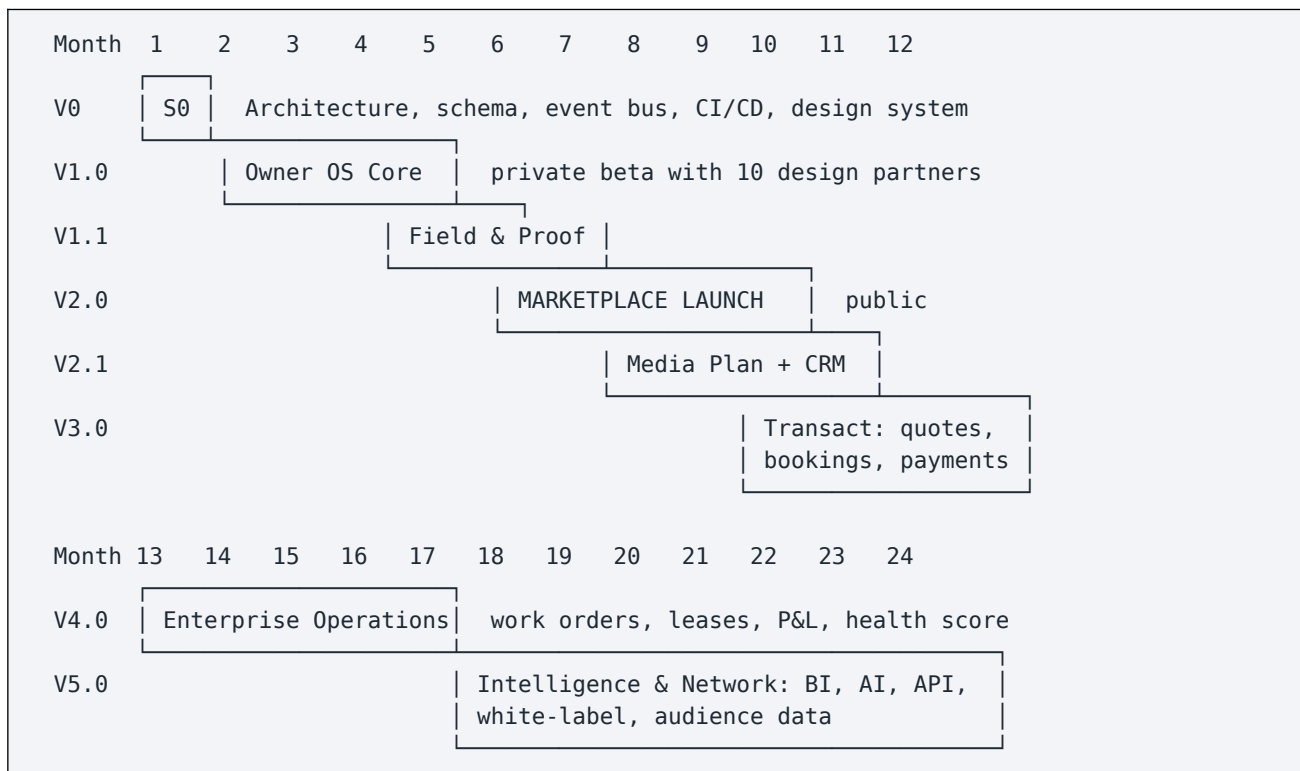
- Four environments: local, development, staging (production-like, with anonymised data), production. No exceptions for "quick fixes".
- Trunk-based development with short-lived feature branches; every merge requires review and green CI.
- Feature flags per tenant so V-next capabilities can be enabled for design partners without a release.
- Weekly deployment cadence minimum; the ability to deploy daily by V2.0.
- Database migrations reviewed by the engineering lead; destructive migrations require a documented rollback plan and a backup checkpoint.

## 16. Release Plan — Version by Version

### 16.1 Release philosophy

18. **Every release must be independently valuable to a real user.** No release exists purely to "prepare" for the next one — except Sprint 0, which is explicitly an architecture investment and is time-boxed to four weeks.
19. **Scope is fixed; dates flex.** When a release is late, we do not remove architectural guardrails, tests or the tenant-isolation suite. We remove **[Should]** items.
20. **Every release has entry criteria and exit criteria.** A release is not "done" when the code is merged; it is done when the exit criteria — which are business outcomes, not tickets — are met.
21. **Design partners before general availability.** Every release is validated with 3–10 named design-partner owners before it is opened to all customers.
22. **The marketplace opens only when the inventory is ready.** This is the single most important sequencing decision in the plan.

### 16.2 Roadmap timeline



### 16.3 V0 — Sprint 0: Architectural Foundation

| Aspect   | Detail                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Duration | Weeks 1–4 (hard time-box)                                                                                                                                                                                                                                                                                                                                                                                                        |
| Goal     | Establish the schema, service boundaries and delivery pipeline so that every later module is additive.                                                                                                                                                                                                                                                                                                                           |
| Scope    | Full domain schema per Section 9 including all reserved entities · multi-tenancy with Row-Level Security · authentication and RBAC skeleton · event bus with publish/subscribe, retries and dead-letter queue · state machine service · polymorphic document service · audit/activity service · CI/CD, environments, observability baseline · design system and component library · API v1 conventions and OpenAPI specification |

| Aspect                        | Detail                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Out of scope                  | Any end-user feature. Sprint 0 produces no customer-visible screens beyond login and a placeholder shell.                                                                                                                                                                                                                                                                                                                                                                                                              |
| Exit criteria                 | (1) A migration suite creating every entity in Section 9.1 runs clean forward and backward. (2) The cross-tenant access test suite passes and is wired into CI as release-blocking. (3) An event published by one service is consumed by two independent subscribers, with a demonstrated retry and dead-letter path. (4) A document can be attached to three different entity types through one API. (5) Staging deploys automatically from the main branch. (6) The OpenAPI specification is published and reviewed. |
| Parallel non-engineering work | WhatsApp BSP application and template drafting started · SMS DLT registration started · design partner recruitment (target 10 signed letters of intent) · onboarding data template (Appendix D) tested with two real owners' spreadsheets · brand identity and domain secured                                                                                                                                                                                                                                          |

## 16.4 V1.0 — Owner OS Core (Private Beta)

| Aspect          | Detail                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Duration        | Months 2–4                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Goal            | Ten media owners stop using their spreadsheet. The permit and contract register lives in HOARDINGS360 and they open it daily.                                                                                                                                                                                                                                                                                                                                |
| Modules         | M01 Identity & tenancy · M02 Asset registry & Board 360 · M03 Document vault · M04 Compliance engine · M05 Clients & agreements · M06 Availability & occupancy engine · M07 Notifications (in-app + email) · M08 Activity feed & audit · M09 Owner dashboard & reports · M10 Bulk import & onboarding · M33 Super admin (basic)                                                                                                                              |
| Out of scope    | Marketplace, mobile app, WhatsApp, incidents, campaigns as an entity (reserved only), payments, work orders, leases                                                                                                                                                                                                                                                                                                                                          |
| Entry criteria  | V0 exit criteria met; 10 design partners signed; at least 3 partners' inventory spreadsheets obtained for import testing                                                                                                                                                                                                                                                                                                                                     |
| Exit criteria   | (1) 10 tenants live with ≥ 1,500 boards digitised in total. (2) ≥ 70% of those boards have GPS, at least one photograph, a rate and at least one compliance record. (3) The vacancy-loop acceptance test in Section 11 (M06) passes in production. (4) ≥ 6 of 10 partners log in at least 3 days per week for 4 consecutive weeks. (5) Zero cross-tenant data incidents. (6) At least 3 partners state in writing that they would pay for the product today. |
| Success metrics | Weekly active tenants · boards digitised · data completeness % · alerts delivered and acknowledged · permit expiries caught before expiry (the headline number for case studies)                                                                                                                                                                                                                                                                             |
| Primary risk    | Data onboarding friction. <b>Mitigation: we do the digitisation for them.</b> A funded onboarding squad visits sites, photographs boards, captures GPS and enters permit data. Treat this cost as customer acquisition, not delivery.                                                                                                                                                                                                                        |

## 16.5 V1.1 — Field, QR & Proof of Display

| Aspect       | Detail                                                                                                                                                |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Duration     | Month 5                                                                                                                                               |
| Goal         | Field teams work from the app, and clients receive photographic proof automatically instead of asking for it.                                         |
| Modules      | M32 H360 Field PWA · M12 Proof of display · M11 Incident management (capture and lifecycle) · M07 extended to WhatsApp and SMS · QR generation in M02 |
| Out of scope | Work order dispatch, SLA engine, technician performance analytics (all V4.0)                                                                          |



| Aspect          | Detail                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Entry criteria  | V1.0 exit criteria met; WhatsApp templates approved; QR stickers produced for at least two partners                                                                                                                                                                                                                                                                             |
| Exit criteria   | (1) $\geq 60\%$ of installations at design-partner sites captured through the app rather than WhatsApp. (2) $\geq 200$ proof captures with valid GPS within tolerance. (3) Offline capture and later sync verified in a genuine low-coverage location. (4) At least 5 client-facing proof packs sent and acknowledged. (5) WhatsApp expiry alerts delivering above 95% success. |
| Success metrics | Proof captures per week · % of campaigns with complete evidence · incidents logged and closed · median incident resolution time (baseline for V4.0 improvement)                                                                                                                                                                                                                 |

## 16.6 V2.0 — Public Marketplace Launch

### This is the highest-risk release in the plan

Everything before this is a controlled beta with friendly users. V2.0 is public, indexed, and judged instantly by professional buyers. **It must not launch early.** The entry criteria below are not aspirational targets — they are gates. If they are not met, the release date moves.

| Aspect                 | Detail                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Duration               | Months 6–8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Goal                   | An advertiser in our launch state can find, evaluate and enquire about real, verified, currently-available inventory without phoning anyone.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Modules                | M25 Map & search · M26 Listing detail & availability calendar · M27 SEO & content engine · M28 Advertiser account, shortlist & enquiry · M29 Hold & reservation · M33 extended (owner verification, listing moderation, content management, platform analytics)                                                                                                                                                                                                                                                                                                                                         |
| Out of scope           | Online payment, instant booking, quotations, campaign portal (all V3.0)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Entry criteria (GATES) | <b>(1) <math>\geq 3,000</math> faces published from <math>\geq 25</math> verified owners. (2) <math>\geq 85\%</math> of published faces have a photograph less than 12 months old, GPS accurate to within 25 m, a rate or rate band, and no expired mandatory permit. (3) Legal pack complete: terms of use, privacy notice, listing policy, DPDP compliance review. (4) Penetration test completed and critical findings closed. (5) Every participating owner has committed in writing to a 24-hour enquiry response SLA. (6) Load test passed at 10<math>\times</math> projected launch traffic.</b> |
| Exit criteria          | (1) $\geq 5,000$ organic marketplace sessions in the launch month. (2) $\geq 150$ verified enquiries submitted. (3) $\geq 60\%$ of enquiries responded to within 24 hours. (4) $\geq 10$ enquiries confirmed by owners as converting to real bookings. (5) $\geq 40$ SEO landing pages indexed and ranking on page 1–2 for their target long-tail terms.                                                                                                                                                                                                                                                |
| Success metrics        | Published faces · verified-owner count · organic sessions · search-to-listing-view rate · listing-view-to-enquiry rate · owner response time · enquiry-to-booking rate (self-reported at this stage)                                                                                                                                                                                                                                                                                                                                                                                                    |
| Primary risk           | Thin or stale inventory. <b>Mitigation: the entry gates, plus a "last verified" date shown publicly on every listing and a quarterly re-verification requirement enforced by the compliance engine.</b>                                                                                                                                                                                                                                                                                                                                                                                                 |

## 16.7 V2.1 — Media Plan Builder & CRM

| Aspect   | Detail                                                                                       |
|----------|----------------------------------------------------------------------------------------------|
| Duration | Month 9                                                                                      |
| Goal     | Collapse proposal creation from hours to minutes and make sure no enquiry or renewal is ever |

| Aspect          | Detail                                                                                                                                                                                                                                                                                                                     |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | dropped.                                                                                                                                                                                                                                                                                                                   |
| Modules         | M15 Media plan & proposal builder · M14 CRM & sales pipeline · M13 Campaign management (full entity and UI)                                                                                                                                                                                                                |
| Entry criteria  | V2.0 exit criteria met; at least 20 real proposals collected from partners to use as design references                                                                                                                                                                                                                     |
| Exit criteria   | (1) Median proposal creation time under 10 minutes, measured in-product. (2) ≥ 100 proposals generated across design partners in the first month. (3) ≥ 50% of proposals shared as interactive links rather than only PDF. (4) Every marketplace enquiry automatically appears as a CRM lead with an owner and a due date. |
| Success metrics | Proposals per tenant per week · proposal-to-campaign conversion rate · enquiry response SLA compliance · renewal follow-ups completed on time                                                                                                                                                                              |
| Marketing note  | <b>This release is the strongest sales demo we will ever have.</b> Build the demo script, the video and the case study alongside the code, not afterwards.                                                                                                                                                                 |

## 16.8 V3.0 — Transact

| Aspect          | Detail                                                                                                                                                                                                                                                                                                                                                                |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Duration        | Months 10–12                                                                                                                                                                                                                                                                                                                                                          |
| Goal            | Money and contracts move through the platform. This is where the marketplace becomes a business rather than a lead-generation channel.                                                                                                                                                                                                                                |
| Modules         | M16 Quotation, booking & contract lifecycle with e-signature · M17 Invoicing, payments & GST · M30 Advertiser campaign portal & proof delivery · M33 extended (billing, commission statements, payouts, disputes)                                                                                                                                                     |
| Entry criteria  | V2.1 exit criteria met; payment gateway merchant onboarding complete; e-signature provider contracted; tax advisor sign-off on GST logic; commission terms accepted by ≥ 20 owners                                                                                                                                                                                    |
| Exit criteria   | (1) ≥ 50 bookings completed end-to-end on the platform. (2) ≥ ₹1 crore cumulative on-platform GMV. (3) ≥ 90% of on-platform bookings invoiced automatically with no manual correction. (4) E-signature completion rate above 80% within 72 hours of sending. (5) Measured leakage rate (enquiries that convert off-platform) established as a baseline and under 40%. |
| Success metrics | GMV · take rate realised · bookings per month · payment success rate · days sales outstanding · leakage rate · advertiser repeat rate                                                                                                                                                                                                                                 |
| Primary risk    | Disintermediation and owner resistance to commission. <b>Mitigation: commission applies only to marketplace-originated demand (Section 5.3); the platform must make on-platform transacting strictly easier through contract generation, e-signature, GST invoicing and proof delivery.</b>                                                                           |

## 16.9 V4.0 — Enterprise Operations

| Aspect   | Detail                                                                                                                                                                                                                                                                 |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Duration | Months 13–16                                                                                                                                                                                                                                                           |
| Goal     | Win Tier A. A 1,000-board agency runs its entire field operation, cost base and asset health on HOARDINGS360.                                                                                                                                                          |
| Modules  | M18 Work order & field force · M19 Landowner lease & expense ledger · M20 Vendor & procurement · M21 Board health score · M22 Financial intelligence & profitability · M04 extended (compliance matrix and score) · M11 extended (SLA, dispatch, technician analytics) |

| Aspect          | Detail                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Entry criteria  | V3.0 exit criteria met; at least 2 Tier A prospects committed as design partners with named executive sponsors                                                                                                                                                                                                                                                                                               |
| Exit criteria   | (1) $\geq 2$ tenants with more than 500 boards fully live. (2) $\geq 1,000$ work orders completed through the platform. (3) Landowner leases and utility costs captured for $\geq 80\%$ of boards at Tier A partners. (4) Per-board profitability reports reconciled against the partner's own accounts to within 5%. (5) Median incident resolution time improved by $\geq 30\%$ against the V1.1 baseline. |
| Success metrics | Enterprise ARR · boards under management · work orders per week · SLA compliance · % of boards with complete cost data · net revenue retention                                                                                                                                                                                                                                                               |

## 16.10 V5.0 — Intelligence & Network

| Aspect          | Detail                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Duration        | Months 17–24                                                                                                                                                                                                                                                                                                                                                                   |
| Goal            | Convert the accumulated inventory graph into products competitors cannot replicate, and open the platform to partners.                                                                                                                                                                                                                                                         |
| Modules         | M23 Business intelligence & heatmaps · M24 AI pricing & vacancy prediction · M31 Audience & impressions data · M34 Public API, webhooks & programmatic supply · M35 White-label & enterprise deployment                                                                                                                                                                        |
| Entry criteria  | V4.0 exit criteria met; $\geq 24$ months of occupancy and pricing history on $\geq 15,000$ faces; opt-in data policy accepted by tenants representing $\geq 60\%$ of inventory                                                                                                                                                                                                 |
| Exit criteria   | (1) Pricing recommendations accepted (used unchanged or lightly adjusted) in $\geq 40\%$ of cases where shown. (2) Vacancy prediction accuracy above an agreed baseline on held-out data. (3) $\geq 3$ partner integrations live on the public API. (4) $\geq 1$ white-label enterprise contract signed. (5) A published market rate benchmark report that the industry cites. |
| Success metrics | API partners and call volume · white-label ARR · benchmark report reach · model acceptance rates · share of national organised inventory on-platform                                                                                                                                                                                                                           |

## 16.11 Release governance

- **Definition of Ready** for any story: user story with acceptance criteria, module ID, designs attached, events identified, permissions identified, test data available.
- **Definition of Done:** code reviewed, unit and integration tests passing, tenant-isolation suite green, events published and verified, audit entries written, permissions enforced, analytics instrumented, documentation updated, deployed to staging and accepted by the Product Owner.
- **Release readiness review** before every version: exit criteria evidence, security review, performance test results, support runbook, rollback plan, customer communication plan.
- **No release without instrumentation.** A feature that cannot be measured against Section 20 does not ship.

## 17. Feature-to-Release Traceability Matrix

The authoritative mapping. Any request to build something not in this table must first be added to it, with a version, by the Product Owner.

| Capability                                   | Module  | V1.0       | V1.1     | V2.0       | V2.1       | V3.0        | V4.0     | V5.0  |
|----------------------------------------------|---------|------------|----------|------------|------------|-------------|----------|-------|
| Multi-tenant accounts, users, RBAC           | M01     | ●          |          |            |            |             | ○        | ○ SSO |
| Board & face registry, Board 360 page        | M02     | ●          | ○ QR     |            |            |             | ○        |       |
| Photograph management, verification dates    | M02     | ●          | ○        | ○          |            |             |          |       |
| Document vault (polymorphic, versioned)      | M03     | ●          |          |            |            | ○ e-sign    | ○ OCR    |       |
| Compliance & permit engine, alerts           | M04     | ●          |          | ○ suppress |            |             | ○ matrix |       |
| Clients & agreements, renewal alerts         | M05     | ●          |          |            | ○ CRM link | ○ e-sign    |          |       |
| Availability & occupancy engine              | M06     | ●          |          | ○ public   |            | ○ bookings  |          |       |
| Auto-listing on becoming vacant              | M06/M25 | ● internal |          | ● public   |            |             |          |       |
| Notifications: in-app + email                | M07     | ●          |          |            |            |             |          |       |
| Notifications: WhatsApp + SMS                | M07     |            | ●        |            |            |             |          |       |
| Activity feed & audit trail                  | M08     | ●          |          |            |            |             |          |       |
| Owner dashboard & standard reports           | M09     | ●          |          |            |            |             | ○ P&L    | ○ BI  |
| Bulk import & guided onboarding              | M10     | ●          | ○ survey |            |            |             |          |       |
| Incident management                          | M11     |            | ●        |            |            |             | ○ SLA    |       |
| Proof of display & proof packs               | M12     |            | ●        |            |            | ○ to portal |          |       |
| Campaign as first-class entity               | M13     | ● reserved |          |            | ●          |             |          |       |
| CRM & sales pipeline                         | M14     |            |          |            | ●          |             |          |       |
| Media plan & proposal builder (PDF/PPT/link) | M15     |            |          |            | ●          |             |          |       |
| Quotations, bookings, agreement generation   | M16     |            |          |            |            | ●           |          |       |
| E-signature                                  | M16     |            |          |            |            | ●           |          |       |
| Invoicing, GST, payments, receivables        | M17     |            |          |            |            | ●           |          |       |
| Commission & payouts                         | M17/M33 |            |          | ○ lead fee |            | ●           |          |       |
| Work orders & field force management         | M18     | ● reserved | ● schema |            |            |             | ●        |       |

| Capability                                  | Module | V1.0       | V1.1       | V2.0     | V2.1 | V3.0      | V4.0  | V5.0         |
|---------------------------------------------|--------|------------|------------|----------|------|-----------|-------|--------------|
| Landowner leases & expense ledger           | M19    | ● reserved |            |          |      |           | ●     |              |
| Vendor & procurement                        | M20    |            | ● reserved |          |      |           | ●     |              |
| Board health score                          | M21    |            |            |          |      |           | ●     |              |
| Profitability per board / district / client | M22    |            |            |          |      |           | ●     |              |
| Business intelligence & heatmaps            | M23    |            |            |          |      |           |       | ●            |
| AI pricing & vacancy prediction             | M24    |            |            |          |      |           |       | ●            |
| Public map & search                         | M25    |            |            | ●        |      |           |       | ○ corr.      |
| Listing detail & availability calendar      | M26    |            |            | ●        |      |           |       |              |
| SEO & content engine                        | M27    |            |            | ●        |      |           |       | ○ multi-lang |
| Advertiser account, shortlist, enquiry      | M28    |            |            | ●        |      | ○ subs    |       |              |
| Hold & reservation                          | M29    |            |            | ●        |      | ○ conv.   |       |              |
| Advertiser campaign portal                  | M30    |            |            |          |      | ●         |       |              |
| Audience & impressions data                 | M31    |            |            |          |      |           |       | ●            |
| Field mobile app (PWA, offline)             | M32    |            | ●          |          |      |           | ○ WOs |              |
| Super admin & platform operations           | M33    | ● basic    |            | ○ verify |      | ○ billing |       |              |
| Public API & webhooks                       | M34    |            |            |          |      |           |       | ●            |
| White-label & enterprise deployment         | M35    |            |            |          |      |           |       | ●            |

**Legend:** ● delivered in this release · ○ materially extended in this release · ● database entity reserved, no user interface

## 18. Key User Journeys & Sample Acceptance Criteria

These ten journeys are the product. They must be covered by automated end-to-end tests and rehearsed before every release. Each is written from the user's point of view, with the release in which it becomes complete.

### 18.1 Journey 1 — Owner onboarding (V1.0)

Sign up → verify email → company profile → download import template  
 → upload spreadsheet → map columns → review validation errors  
 → fix and re-upload → import 240 boards → bulk-upload photographs  
 → onboarding tracker shows 62% complete → add permits for top 50 boards  
 → first compliance alert received within 24 hours

**Success:** an owner with 240 boards reaches 60%+ data completeness within 5 working days, with no more than 2 hours of their own time invested.

### 18.2 Journey 2 — The vacancy loop (V1.0 internal, V2.0 public) — our signature journey

Agreement on face BLR-HSR-0142-A ends 30 September  
 → 60 days out: sales executive alerted, CRM renewal task created  
 → 30 days out: face auto-published to marketplace "Available from 01 Oct"  
     (current campaign remains live and untouched)  
 → advertiser finds it on the map, filters by "available in October"  
 → enquiry submitted → owner responds → hold placed → booked for 05 Oct  
 → face never goes vacant; no self-promotional flex board is ever printed

#### Acceptance criteria (V1.0)

- **Given** a tenant pre-listing window of 30 days and an agreement ending 30 September, **when** the date reaches 31 August, **then** `BOARD\_BECOMING\_VACANT` is emitted exactly once, the face is flagged as available from 01 October, and the assigned sales executive receives an alert on their configured channels.
- **Given** the same face, **when** the contract is extended before expiry, **then** the pre-listing is withdrawn automatically, a `LISTING\_UNPUBLISHED` event is emitted, and no advertiser who had shortlisted it can submit an enquiry for a conflicting period.
- **Given** the face has an expired municipal licence, **when** the pre-listing window is reached, **then** the face is **not** published, and a compliance alert is raised instead.

### 18.3 Journey 3 — Permit renewal (V1.0)

90 days before expiry → compliance officer alerted (email + in-app)  
 → marks record "under renewal", records application date  
 → 60 / 30 / 15 / 7 day reminders continue until resolved  
 → uploads new certificate → old record superseded, new expiry computed  
 → PERMIT\_RENEWED → board compliance status returns to valid  
 → if expiry passes with no renewal: PERMIT\_EXPIRED → face unpublished,  
     admin + compliance officer alerted at high priority, daily until resolved

### 18.4 Journey 4 — Advertiser discovery to enquiry (V2.0)

Google: "hoarding advertising cost in Kochi" → SEO landing page  
 → map with 180 available faces → filter: unipole, lit, October, ₹40k–80k  
 → 23 results → open listing → photographs, specs, availability calendar  
 → add 6 boards to shortlist → register with OTP → submit one enquiry  
 → split automatically into 4 owner-specific enquiries

- owners notified on WhatsApp; SLA clock starts
- responses with rates arrive in advertiser inbox within 24 hours

#### Acceptance criteria

- **Given** a shortlist spanning four owners, **when** one enquiry is submitted, **then** four separate enquiries are created and no owner can see another owner's boards, rates or the advertiser's full shortlist.
- **Given** an enquiry unanswered for 24 hours, **when** the SLA elapses, **then** the owner's company admin is notified and the enquiry is flagged in platform analytics.

### 18.5 Journey 5 — Proposal in ten minutes (V2.1)

Brief received: "20 boards, Kochi + Thrissur, November, ₹15 lakh"

- open map, filter by city + availability + budget
- select 20 faces → create media plan → apply 8% volume discount
- generate branded PDF + PPT → share interactive link with client
- PROPOSAL\_VIEWED notification → client rejects 3 sites, approves 17
- one click: convert to campaign → (V3.0) generate quotation → e-sign

### 18.6 Journey 6 — Installation and proof (V1.1)

Campaign starts 01 November → installation task appears for the crew

- technician opens H360 Field → today's jobs → navigate
- scans QR on the structure → confirms correct board
- captures before photo → installs flex → captures day photo, night photo, approach photo (GPS + timestamp embedded)
- offline? captures queue and sync on return to coverage
- PROOF\_UPLOADED → proof pack generated → (V3.0) appears instantly in the advertiser's campaign portal

### 18.7 Journey 7 — Incident to resolution (V1.1 capture, V4.0 dispatch)

Client reports the light is out on board 142 (or public QR report)

- incident created, category Electrical / lighting failure, severity high
- face is occupied → account manager notified proactively
- work order auto-generated with 24-hour SLA (V4.0)
- assigned to the electrician covering that zone
- technician checks in on site (GPS), repairs, uploads after-photo
- supervisor verifies → incident closed → health score recalculated
- client notified with evidence, without ever having to chase

### 18.8 Journey 8 — Booking and payment (V3.0)

Advertiser accepts quotation → booking created → faces locked

- agreement generated from template → sent for e-signature
- advertiser signs on mobile → signed PDF stored in vault
- invoice issued with correct GST treatment → payment link sent
- payment received → BOOKING\_CONFIRMED → occupancy updated
- installation work orders generated → commission statement to owner

### 18.9 Journey 9 — Month-end for an operations manager (V4.0)

Open dashboard → 1,040 faces, 86% occupied, 14 permits expiring in 30 days

- 9 open incidents, 2 breaching SLA → reassign from the scheduling board
- vacancy report: 12 faces vacant > 30 days, ₹8.4 lakh foregone
- profitability: 7 boards net-negative after land rent and electricity
- decision: renegotiate 3 leases, retire 2 structures, reprice 2
- landowner payments due this week: 22, all scheduled

## 18.10 Journey 10 — Platform onboarding a new owner (V2.0, Super Admin)

- Owner signs up → uploads company registration, GST, sample permits, site ownership or lease evidence
- platform reviewer checks documents and spot-checks 5 board locations
  - duplicate detection: 3 boards already claimed by another tenant
  - flagged for manual resolution before publication
  - approved → "Verified" badge issued → listings go live
  - 90 days later: re-verification prompt if photographs are stale

## 18.11 Sample user stories for Sprint 1

| ID       | Story                                                                                                                                        | Key acceptance criteria                                                                                                                                                                                |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| H360-101 | As a company admin, I can register my company and invite my team with specific roles, so that each person sees only what their job requires. | Invitation email delivered; role assigned at invitation; invited user cannot access any module outside their permission set; every role change appears in the audit log.                               |
| H360-102 | As an operations manager, I can add a structure with multiple faces, so that a two-sided unipole is sold as two independent units.           | Faces have independent dimensions, illumination, orientation and rates; occupancy is tracked per face; the Board 360 page lists all faces with their individual status.                                |
| H360-103 | As a compliance officer, I can record several different clearances on one board, each with its own authority and expiry.                     | At least five clearance types supported and configurable; each generates independent alerts; board compliance status rolls up correctly; expired clearance blocks marketplace publication.             |
| H360-104 | As a sales executive, I want to be alerted before a client contract ends, so that I can start renewal conversations early.                   | Alerts at 90/60/30/15/7 days (configurable); delivered to the assigned executive; acknowledgement recorded; CRM task created from V2.1.                                                                |
| H360-105 | As a media owner, I can import 500 boards from my existing spreadsheet in one operation.                                                     | Template download; column mapping; per-row validation with clear messages; valid rows commit while invalid rows are returned for correction; duplicates within 25 m flagged; summary report generated. |



## 19. Go-to-Market Plan

### The one strategic instruction for the commercial team

We are not selling software. We are **recruiting a network**, and the software is how we keep it. For the first twelve months, every marketing rupee and every sales hour is measured against one number: **verified faces on the platform, kept fresh**. Revenue in year one is a secondary objective; inventory density is the primary one, because it is the only thing that makes year two possible.

### 19.1 Sequencing: why supply comes first

A two-sided marketplace can be started from either side, but in OOH the supply side is the harder, slower and more defensible one. Advertisers can be acquired with search marketing in weeks. Media owners take months of relationship-building, and once their permit register and contract history live in our system, they do not leave. **Therefore: build supply density in a narrow geography, then switch on demand.**

| Phase                     | Window                  | Objective                                  | Target                                    | Primary activity                                                                 |
|---------------------------|-------------------------|--------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------|
| P1 — Design partners      | Months 1–4 (V0–V1.0)    | Prove the ERP is worth using daily         | 10 owners · 1,500 boards                  | Founder-led selling; free digitisation; weekly feedback calls                    |
| P2 — Depth in one state   | Months 5–8 (V1.1–V2.0)  | Reach marketplace-viable density in Kerala | 25–40 owners · 3,000+ faces               | Field sales team of 2; district-by-district coverage; trade association presence |
| P3 — Demand switch-on     | Months 6–9 (V2.0)       | Prove advertisers arrive and convert       | 5,000 sessions/mo · 150 enquiries/mo      | SEO engine live; agency outreach; PR around the launch                           |
| P4 — Regional expansion   | Months 9–16 (V2.1–V4.0) | Replicate the Kerala playbook              | 250 owners · 25,000 faces across 4 states | Repeatable playbook; inside sales + field; partner channel                       |
| P5 — Enterprise & network | Months 17–24 (V5.0)     | Land Tier A and open the API               | 5 Tier A logos · 3 API partners           | Enterprise sales motion; industry data products; category leadership             |

### Why Kerala first

- **Manageable geography.** A field team can physically reach any board in the state within a day, which matters enormously for the digitisation squad.
- **Dense, organised ownership.** Enough mid-size owners to reach marketplace density without needing a national footprint.
- **Contained competitive attention.** National players concentrate on Mumbai, Delhi and Bengaluru; we can build a credible network before anyone reacts.
- **Provable playbook.** Owning one state completely is a far stronger investor and customer story than a thin presence in six.
- **Expansion order thereafter:** Tamil Nadu → Karnataka → Telangana/Andhra Pradesh → Maharashtra & Gujarat, entering each state only when the prior one exceeds 70% of identified organised inventory.

### 19.2 The Founding Partner Program (supply acquisition engine)

A time-boxed, deliberately generous programme for the first 25 media owners in each new state. **It is the only sanctioned discount mechanism in the business.**

| We give                                                                                                                                          | They give                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Free digitisation of up to 150 boards: our team visits sites, photographs day and night, captures GPS, records dimensions and enters permit data | Access to their inventory records and permit files, and site access for our surveyors                      |
| 12 months free on the Professional plan, then 40% off for life                                                                                   | A signed commitment to keep availability data current and respond to marketplace enquiries within 24 hours |
| Priority placement on the marketplace for the first 6 months                                                                                     | A written testimonial and agreement to take up to 4 reference calls per year                               |
| Direct line to the product team and named influence over the roadmap                                                                             | A monthly 30-minute feedback call during the design-partner period                                         |
| Co-marketing: joint case study, PR mention, logo on the site                                                                                     | Permission to use their name and results publicly                                                          |

**Discipline:** the programme closes at 25 owners per state or at the state's marketplace launch, whichever comes first. Late entrants pay list price — and that scarcity is itself a closing argument for the early ones.

## 19.3 Supply acquisition playbook

### Target list construction

- Build a physical inventory census: drive the primary corridors of each target city and photograph every structure. Owner contact numbers are, conveniently, printed on the vacant ones. **This census is simultaneously a sales list, a market-size validation and the seed of the marketplace database.**
- Cross-reference with municipal advertising-licence registers where they are published, trade association membership lists, and existing agency relationships.
- Segment by Section 4.1 tiers; prioritise Tier B owners with 50–500 boards in the launch geography.

### The sales conversation

23. **Open with their risk, not our features.** "How many of your municipal licences expire in the next 90 days? Can you tell me in under a minute?" Almost no one can.
24. **Show their own boards.** Before the meeting, import 20–30 of their structures from the census. Nothing else in the demo comes close to this for impact.
25. **Quantify the vacancy loss.** "Board 142 was empty 47 days last year. At your card rate that is ₹1.4 lakh. Our annual fee is ₹1.2 lakh." The arithmetic closes the deal, not the feature list.
26. **Handle the data objection immediately.** "You will not type anything. We will come and do it." Digitisation-as-onboarding removes the single biggest adoption barrier.
27. **Position the marketplace as upside, not the pitch.** If they buy for the marketplace and it is early, they churn. If they buy for compliance and vacancy, the marketplace is a bonus that keeps them.

### Objection handling

| Objection                                                  | Response                                                                                                                                                                                                                                           |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "My competitors will see my inventory and rates."          | Rate display is per-tenant configurable — you can show a band or "on request". Your cost data, floor rates, client names and contract values are never public. And your competitors already learn your rates from the same clients you both pitch. |
| "I already use Excel and it works fine."                   | Excel does not send you a WhatsApp message 90 days before a licence expires, and it does not sell your board while the current campaign is still running. Which of your 240 boards is free next month? — the pause before the answer is the pitch. |
| "My data will be misused / you will become my competitor." | We publish a written neutrality policy: HOARDINGS360 never owns, leases or resells inventory, and never acts as a media buyer. Aggregate benchmarking is strictly opt-in and anonymised.                                                           |

| Objection                              | Response                                                                                                                                                                                                       |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "My staff will not use software."      | The field app is a QR scan and two photographs. Your technicians already use WhatsApp; this is simpler and it proves they did the job.                                                                         |
| "Too expensive."                       | One missed permit renewal or one 30-day vacancy costs more than a year of subscription. We will show you the number from your own boards. Price is not discounted; annual prepay and digitisation support are. |
| "Come back when you have advertisers." | That is exactly why the first 25 owners get the Founding Partner terms — you are being paid, in free digitisation and lifetime pricing, to be early. In six months these terms are gone.                       |

## 19.4 Demand generation (advertiser side)

### Channel priorities

| Channel                                     | Role          | Why                                                                                                                                         | Starts                                   |
|---------------------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Organic search / programmatic SEO (M27)     | Primary       | Buyers search "hoarding rates in <city>" months before they call anyone. Compounding, defensible, and our inventory database is the content | Content from Month 4; pages live at V2.0 |
| Agency & brand direct outreach              | Primary       | Agencies place the large, repeatable volume. One planner relationship is worth a thousand SME visits                                        | Month 6                                  |
| Paid search                                 | Supporting    | Fill the gap while SEO matures; tightly bounded budget on high-intent commercial terms only                                                 | V2.0 launch                              |
| Industry PR & trade media                   | Supporting    | Credibility with owners as much as buyers; the launch of a state-wide inventory map is a genuine trade story                                | V2.0 launch                              |
| LinkedIn content & founder voice            | Supporting    | Reaches marketing heads and agency planners; builds category authority cheaply                                                              | Month 3                                  |
| Regional business associations & SME bodies | Experimental  | Local advertisers (retail chains, hospitals, education, real estate) are underserved and price-sensitive                                    | V2.1                                     |
| Political and event seasonality campaigns   | Opportunistic | Election and festival cycles create sharp, short-term OOH demand spikes — plan inventory and campaigns around the calendar                  | Ongoing                                  |

### SEO content plan (jointly owned by Product and Marketing)

- **Programmatic pages:** city, locality, highway, format, and format × city — each with live inventory counts, real price ranges, a map and sample sites.
- **Rate guides:** "Hoarding advertising cost in Kochi 2026" style pages, updated from actual platform data. These are the highest-intent queries in the category and we will have better data than anyone publishing them today.
- **Explainers:** unipole versus billboard versus gantry; front-lit versus back-lit; how municipal permissions work; what a media plan should contain.
- **Case studies:** campaign stories with proof-of-display photographs — content no competitor can fabricate.
- **Discipline:** every page must contain data a human would not otherwise have. Thin programmatic pages will be de-indexed and will damage the whole domain.

## 19.5 Brand, positioning & messaging

### Brand architecture

| Brand        | Applies to                              | Voice                                                                                       |
|--------------|-----------------------------------------|---------------------------------------------------------------------------------------------|
| HOARDINGS360 | Master brand and the public marketplace | Confident, transparent, modern — the trustworthy front door to outdoor advertising in India |
| H360 Manage  | Owner platform                          | Operational, precise, reassuring — "nothing slips"                                          |
| H360 Field   | Mobile field app                        | Simple, direct, respectful of the technician's time                                         |

**Naming caution:** "HOARDINGS360" is descriptive and works well in India, but reads as narrow if the product later covers transit, retail and airport media, and is weak in international search. **Decision required in Sprint 0 (Section 23, OQ-8):** keep it as the master brand, or adopt a broader parent brand with HOARDINGS360 retained as the marketplace property. Whatever is chosen, secure the domain, trademark class 9/35/42 filings and social handles before any public launch.

### Messaging house

| Audience                    | Core promise                                               | Supporting proof points                                                                                                          | Line to use                                                              |
|-----------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Media owner (Rajesh)        | Never lose a permit, never lose a day of occupancy         | Multi-clearance compliance engine · 90/60/30/15/7-day alerts · auto-listing 30 days before a contract ends · vacancy loss report | "Know every board. Sell every day."                                      |
| Operations manager (Priya)  | Every job tracked, every campaign proved                   | Incidents · work orders · GPS check-in · geo-tagged proof packs                                                                  | "Stop running your operation on WhatsApp."                               |
| Sales executive (Arun)      | Proposals in minutes, not hours                            | Map-based selection · branded PDF/PPT · interactive client link · live availability                                              | "Twenty boards to a client-ready proposal in ten minutes."               |
| Advertiser / agency (Meera) | See every available board, with real dates and real prices | Verified listings · availability calendars · photographs with capture dates · proof of display                                   | "Plan your outdoor campaign without a single phone call."                |
| Investor / stakeholder      | The system of record becomes the marketplace               | Supply-first sequencing · operational stickiness · inventory graph as moat                                                       | "We are digitising India's outdoor inventory — and then transacting it." |

### Proof assets to build alongside the product

- **The vacancy-loop demo video** (90 seconds): contract ending → auto-listing → enquiry → booked before it goes empty. Build this at V2.0; it is the single most persuasive asset we will own.
- **The compliance risk report:** a free, personalised audit of a prospect's licence exposure, generated from the census data. A lead magnet that is also a demo.
- **Case study template:** boards digitised, vacancy days reduced, permits caught, proposal time saved, revenue recovered. Collect these numbers from design partners from V1.0 onward — retrofitting them later is impossible.
- **Live inventory counter** on the marketplace ("4,180 boards · 38 verified owners · 6 cities"), which simultaneously builds buyer confidence and applies competitive pressure on owners not yet listed.

## 19.6 Launch calendar by release

| Release       | Marketing & commercial activity                                                                          |
|---------------|----------------------------------------------------------------------------------------------------------|
| V0 (Sprint 0) | Brand identity, domain and trademark filings · design partner recruitment (10 LOIs) · physical inventory |

| Release | Marketing & commercial activity                                                                                                                                                                               |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | census of the launch city begins · founding-partner terms finalised                                                                                                                                           |
| V1.0    | Private, no public launch · weekly design partner calls · begin collecting case study baseline numbers · start SEO content production (not yet published) · build sales demo environment with real board data |
| V1.1    | First case study drafted · proof-of-display becomes a demo centrepiece · QR stickers deployed as a physical branding surface on partner boards                                                                |
| V2.0    | <b>Public launch.</b> Trade PR · launch event with an industry body · SEO pages live · paid search switched on · agency outreach begins · live inventory counter published · founding partner logos featured  |
| V2.1    | Sales-led push using the proposal builder demo · webinar for owners: "cut your proposal time by 90%" · second case study                                                                                      |
| V3.0    | Advertiser acquisition campaign with payment protection and proof-of-display as the trust message · first GMV milestone PR · commission model communicated to owners with a 60-day notice period              |
| V4.0    | Enterprise motion begins: whitepaper on OOH operational efficiency · Tier A account-based marketing · industry conference presence                                                                            |
| V5.0    | Category leadership: publish the first India OOH occupancy and rate benchmark report · API partner programme launch · white-label sales motion                                                                |

## 19.7 Customer success & retention

- **Onboarding is the retention strategy.** A tenant that reaches 70% data completeness in the first 30 days is dramatically more likely to still be active at month twelve. Staff and measure onboarding accordingly.
- **Health scoring for accounts:** weekly active users, data completeness, alerts acknowledged, enquiries answered, boards added in the last 90 days. Intervene when the score falls, not when the renewal date arrives.
- **Quarterly business review** for Professional and above: occupancy trend, vacancy recovered, permits caught, proposals generated, enquiries received. **Show them the value in their own numbers — this is what makes renewal a formality.**
- **A named onboarding manager** for every Founding Partner. This is expensive and correct.
- **Churn post-mortem** on every lost account, reviewed by the Product Owner monthly. In a system of record, churn is almost always an onboarding failure, not a feature gap.

## 19.8 Marketing KPIs

| Stage      | KPI                                | V2.0 target | V3.0 target |
|------------|------------------------------------|-------------|-------------|
| Supply     | Verified owners onboarded          | 25–40       | 80          |
| Supply     | Published faces                    | 3,000       | 10,000      |
| Supply     | Data freshness (photo < 12 months) | ≥ 85%       | ≥ 90%       |
| Demand     | Monthly organic sessions           | 5,000       | 30,000      |
| Demand     | Monthly verified enquiries         | 150         | 600         |
| Demand     | Listing-view to enquiry rate       | ≥ 3%        | ≥ 4%        |
| Conversion | Owner response within 24 h         | ≥ 60%       | ≥ 85%       |
| Conversion | Enquiry to booking                 | measured    | ≥ 12%       |
| Commercial | Paying tenants                     | 15          | 60          |
| Commercial | On-platform GMV (cumulative)       | —           | ₹1 crore    |

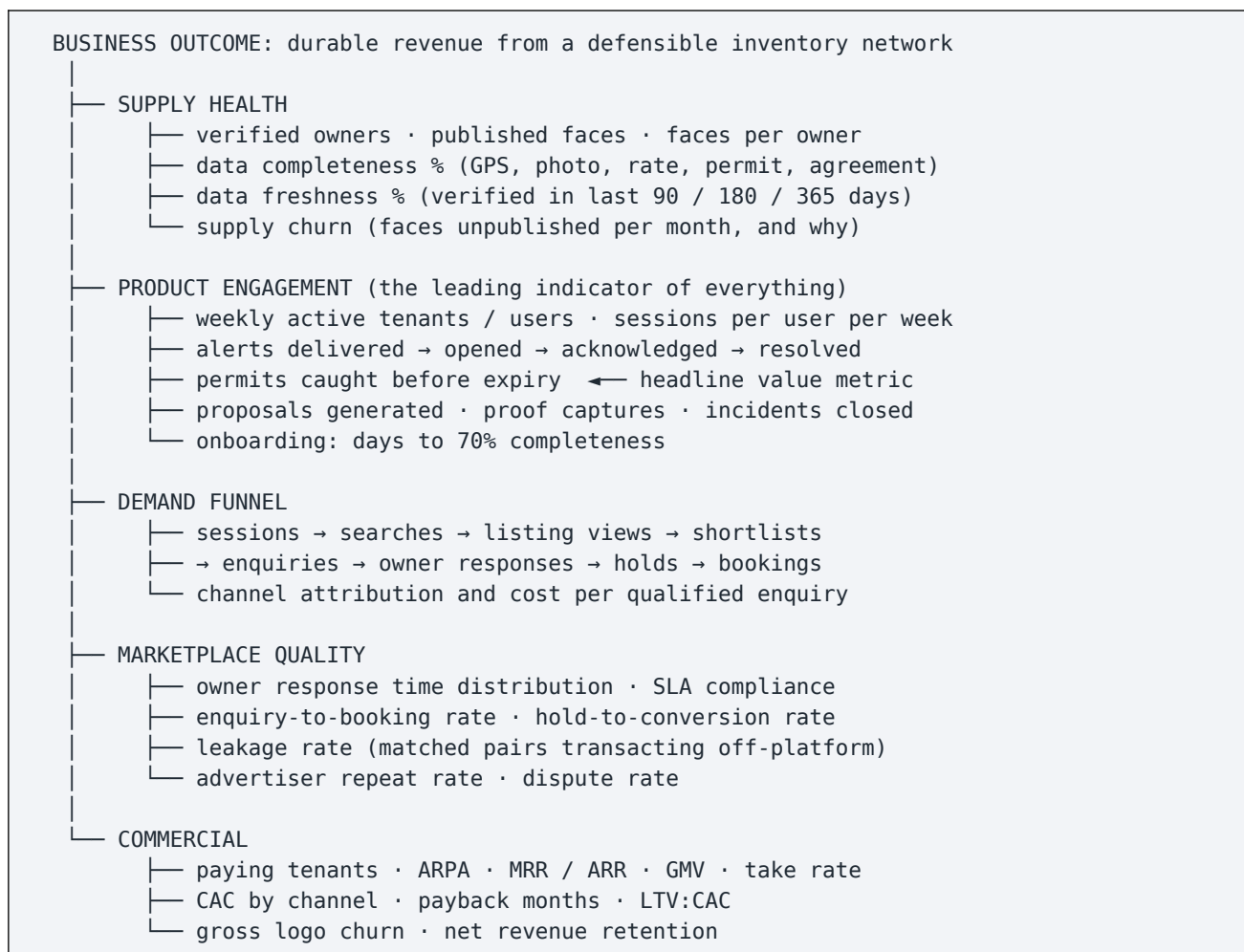


## 20. Metrics Framework & Instrumentation

### 20.1 North Star metric by phase

| Phase     | North Star                                                                                                                                    | Why this one                                                                                          |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| V1.0-V1.1 | <b>Actively Managed Faces</b> — faces with complete data that were touched (viewed, updated, alerted-on and acknowledged) in the last 30 days | Measures real adoption, not sign-ups. A digitised board nobody looks at is worthless                  |
| V2.0-V2.1 | <b>Qualified Enquiries Answered within SLA</b>                                                                                                | Captures both sides at once: enough inventory to attract demand, and owners engaged enough to respond |
| V3.0+     | <b>On-platform Booked Face-Months</b>                                                                                                         | The unit of value actually transacted; converts directly into GMV and commission                      |
| V4.0-V5.0 | <b>Net Revenue Retention</b>                                                                                                                  | At maturity, the question is whether accounts expand. NRR is the truest measure of a system of record |

### 20.2 KPI tree



### 20.3 Instrumentation requirements

- **Every event in Section 10.1 is also an analytics event**, with tenant, user, role, entity and timestamp. Build this once in the event consumer; do not instrument screens individually.

- Front-end product analytics on all three surfaces from V1.0, with a documented event naming convention agreed in Sprint 0.
- **Vacancy accounting** must be exact: for every face, days vacant, reason (contract ended, permit expired, maintenance, blocked), and card-rate value foregone. This single dataset powers the sales pitch, the case studies and the product's own value narrative.
- **Leakage measurement from V3.0:** track advertiser-owner pairs that exchange contact details through an enquiry and then show no on-platform booking within 90 days. Follow up by survey. **A rising leakage rate is a product problem to be fixed with utility, not a policing problem.**
- Cohort analysis by onboarding month for tenants and by first-session month for advertisers.

## 20.4 Reporting cadence

| Cadence     | Audience                         | Content                                                                                        |
|-------------|----------------------------------|------------------------------------------------------------------------------------------------|
| Daily       | Product & engineering            | Error rates, queue depth, notification failures, failed imports, uptime                        |
| Weekly      | Product Owner + commercial leads | North Star, supply growth, funnel, active tenants, onboarding pipeline, top 5 support themes   |
| Monthly     | Founders & steering group        | Full KPI tree, cohort retention, CAC/payback, churn post-mortems, roadmap variance             |
| Per release | All stakeholders                 | Exit criteria evidence, adoption of shipped features 30 days in, defects escaped to production |
| Quarterly   | Board / investors                | Strategy review, market position, competitive movement, re-baselined plan                      |



## 21. Risk Register

Scored as Probability (P) × Impact (I), each 1–5. Owner is a named function, not a committee. Reviewed monthly; any risk scoring ≥ 16 is escalated to the steering group with a written mitigation plan.

| ID  | Risk                                                                                                       | P | I | Score | Mitigation                                                                                                                                                          | Owner                    |
|-----|------------------------------------------------------------------------------------------------------------|---|---|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| R1  | <b>Cold start:</b> marketplace launches with too little inventory and loses buyer credibility permanently  | 4 | 5 | 20    | Hard entry gates at V2.0 (Section 16.6); Founding Partner Program; state-by-state density before expansion; live inventory counter only published once credible     | Founder / GTM            |
| R2  | <b>Data onboarding friction:</b> owners will not enter their inventory                                     | 5 | 4 | 20    | Bulk import (M10); funded digitisation squad; field survey mode; onboarding tracker; treat digitisation as CAC                                                      | Customer Success         |
| R3  | <b>Architecture rewrite:</b> V4.0 modules cannot be added without restructuring the database               | 3 | 5 | 15    | Section 8 guardrails; reserved entities in Sprint 0; architecture review gate before every release                                                                  | Engineering Lead         |
| R4  | <b>Disintermediation:</b> owners and advertisers transact off-platform                                     | 5 | 3 | 15    | Commission only on marketplace-originated demand; make on-platform transacting easier (contracts, e-sign, GST, proof); measure leakage from V3.0                    | Product Owner            |
| R5  | <b>Data trust:</b> owners fear exposing rates and inventory to competitors                                 | 4 | 4 | 16    | Per-tenant pricing visibility policy; published neutrality policy; cost and client data never public; opt-in only for benchmarks                                    | Founder                  |
| R6  | <b>Stale listings</b> destroy marketplace credibility over time                                            | 4 | 4 | 16    | "Last verified" date public on every listing; quarterly re-verification enforced; ranking penalty for stale inventory; auto-suppression on expired permits          | Product Owner            |
| R7  | <b>Regulatory/compliance exposure:</b> we publish inventory that is later found to be unpermitted          | 3 | 4 | 12    | Permit-expired faces auto-unpublished; verification at onboarding; terms placing responsibility with the owner; legal review before V2.0                            | Legal / Founder          |
| R8  | <b>DPDP and data protection non-compliance</b>                                                             | 3 | 4 | 12    | Counsel engaged before V2.0; consent notices; data-principal rights implemented; residency in India; documented retention                                           | Legal / Engineering Lead |
| R9  | <b>Key integration delay:</b> WhatsApp template approval, DLT registration or payment KYC blocks a release | 4 | 3 | 12    | Start all provider onboarding in Sprint 1; design channel-agnostic notification service so email works standalone                                                   | Engineering Lead         |
| R10 | <b>Incumbent response:</b> an established software vendor bolts on a marketplace                           | 3 | 4 | 12    | Compete on compliance and landowner depth; move fast on state density; lock in multi-year Founding Partner terms                                                    | Founder                  |
| R11 | <b>Agency channel conflict:</b> agencies see us as a threat to their margin                                | 3 | 3 | 9     | Position as a planning tool that saves them time; offer agency accounts and multi-client workspaces; never sell direct against a partner agency on their own client | GTM Lead                 |
| R12 | <b>Scope creep</b> from the enterprise wish-list pulling V1.0 out of                                       | 4 | 3 | 12    | Traceability matrix (Section 17); Product Owner is the sole gate; reserved-schema pattern lets                                                                      | Product Owner            |

| ID  | Risk                                                                    | P | I | Score | Mitigation                                                                                                                      | Owner            |
|-----|-------------------------------------------------------------------------|---|---|-------|---------------------------------------------------------------------------------------------------------------------------------|------------------|
|     | shape                                                                   |   |   |       | us say "designed for, not built yet"                                                                                            |                  |
| R13 | <b>Field app non-adoption</b> by technicians                            | 3 | 3 | 9     | Extreme simplicity; QR-first; offline tolerance; regional language; involve technicians in usability testing, not just managers | Product Owner    |
| R14 | <b>Duplicate/contested boards:</b> two tenants claim the same structure | 4 | 2 | 8     | Proximity duplicate detection at import and publication; super-admin resolution workflow; ownership evidence at verification    | Platform Ops     |
| R15 | <b>Concentration risk:</b> a few large tenants dominate revenue         | 3 | 3 | 9     | Maintain Tier B breadth; cap any single tenant above an agreed revenue share in planning assumptions                            | Founder          |
| R16 | <b>Talent and key-person risk</b> in a small engineering team           | 3 | 4 | 12    | Documentation as a Definition of Done item; no single-owner modules; pair on the compliance and occupancy engines specifically  | Engineering Lead |
| R17 | <b>Photograph storage cost</b> growth outpaces revenue                  | 3 | 2 | 6     | Compression on capture; lifecycle tiering; per-plan storage limits; CDN caching                                                 | Engineering Lead |

## 22. Team, Governance & Ways of Working

### 22.1 Team composition by phase

| Role                            | V0-V1.1     | V2.0-V3.0 | V4.0-V5.0 | Notes                                                                       |
|---------------------------------|-------------|-----------|-----------|-----------------------------------------------------------------------------|
| Product Owner                   | 1           | 1         | 1-2       | Single accountable decision-maker on scope. Non-negotiable role             |
| Engineering Lead / Architect    | 1           | 1         | 1         | Owens Section 8 guardrails and every schema change                          |
| Backend engineers               | 2           | 3         | 4         | TypeScript/Node or Python; one must own the geospatial and occupancy engine |
| Frontend engineers              | 2           | 3         | 4         | One dedicated to the marketplace (SSR/SEO) from V2.0                        |
| Mobile/PWA engineer             | 0.5         | 1         | 1         | Can be a frontend engineer through V1.1                                     |
| UI/UX designer                  | 1           | 1         | 1-2       | Design system in Sprint 0; field app usability testing on real devices      |
| QA engineer                     | 0.5         | 1         | 2         | Owens the tenant-isolation suite and end-to-end journeys                    |
| DevOps / SRE                    | 0.5         | 1         | 1         | Part-time acceptable to V2.0, not beyond                                    |
| Data/BI engineer                | 0           | 0.5       | 1         | From V3.0 for the warehouse; V5.0 for models                                |
| Onboarding & digitisation squad | 2           | 4         | 6+        | <b>Field staff, not engineers. Budget as customer acquisition</b>           |
| Sales (field + inside)          | 1 (founder) | 3         | 6         | Founder-led until V2.0                                                      |
| Marketing / content             | 0.5         | 1.5       | 2         | SEO content production must start at Month 4                                |
| Customer success                | 0           | 1         | 2-3       | From the first paying cohort                                                |

### 22.2 Ways of working

- Two-week sprints with a demo to at least one design partner at the end of every sprint. **A demo to internal stakeholders only does not count.**
- The Product Owner writes or approves every user story; engineering may not add scope without a matrix entry (Section 17).
- An architecture review gate before each release, chaired by the Engineering Lead, explicitly checking the ten rules in Section 8.
- A written decision log: every material product, architecture or commercial decision recorded with date, options considered, decision and rationale. **This document is the baseline; the decision log is its running amendment.**
- Monthly steering group: founders, Product Owner, Engineering Lead, GTM lead. Reviews the KPI tree, the risk register and roadmap variance.
- Support rota from V1.0, with a published response commitment to design partners. Early support quality is a product feature.

## 22.3 RACI for key decisions

| Decision                               | Responsible      | Accountable      | Consulted               | Informed         |
|----------------------------------------|------------------|------------------|-------------------------|------------------|
| Release scope and priority             | Product Owner    | Founder          | Eng Lead, GTM, Design   | All              |
| Architecture and stack choices         | Engineering Lead | Engineering Lead | Product Owner, DevOps   | Founder          |
| Pricing and packaging                  | GTM Lead         | Founder          | Product Owner, Finance  | All              |
| Marketplace launch go/no-go            | Product Owner    | Founder          | Eng Lead, GTM, Legal    | All              |
| Founding Partner terms                 | GTM Lead         | Founder          | Finance                 | Product, Success |
| Data, privacy and neutrality policy    | Legal counsel    | Founder          | Eng Lead, Product Owner | All              |
| Deferring a feature to a later version | Product Owner    | Product Owner    | Eng Lead, GTM           | All              |

## 22.4 Indicative effort and budget shape

Directional only; to be replaced with a costed plan by the Engineering Lead and Finance in Sprint 0.

| Release       | Engineering effort (person-months) | Dominant non-engineering cost                                         |
|---------------|------------------------------------|-----------------------------------------------------------------------|
| V0 — Sprint 0 | 6–8                                | Design system, legal setup, brand                                     |
| V1.0          | 18–24                              | <b>Digitisation squad</b> (the largest single line in the early plan) |
| V1.1          | 6–8                                | WhatsApp/SMS provider fees, QR sticker production                     |
| V2.0          | 20–26                              | SEO content production, PR, penetration test, launch event            |
| V2.1          | 8–10                               | Sales enablement, demo video, webinars                                |
| V3.0          | 20–26                              | Payment gateway and e-sign fees, tax and legal advisory               |
| V4.0          | 28–36                              | Enterprise sales motion, implementation support                       |
| V5.0          | 30–40                              | Data partnerships, benchmark report, developer relations              |

## 23. Open Decisions & Assumptions

### 23.1 Decisions required before or during Sprint 0

| ID    | Question                                                                                                                                                                  | Owner                    | Needed by                       | Default if undecided                                                                    |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------------------------------|-----------------------------------------------------------------------------------------|
| OQ-1  | Bottom-up market sizing: how many organised structures exist in the launch state, and what is realised rent per board? (Validate with the census and the first 10 owners) | Founder / GTM            | Month 3                         | Proceed on published range; do not fundraise on it                                      |
| OQ-2  | Launch geography confirmation: Kerala first, or a different state where founder relationships are strongest?                                                              | Founder                  | Sprint 0                        | Kerala                                                                                  |
| OQ-3  | Backend language: NestJS/TypeScript or Python/FastAPI?                                                                                                                    | Engineering Lead         | Sprint 0 week 1                 | NestJS/TypeScript (single-language stack)                                               |
| OQ-4  | Mapping provider: Google Maps Platform or Mapbox? Evaluate on India data quality, clustering at 25k+ markers, and cost curve                                              | Engineering Lead         | Sprint 0                        | Google Maps for data quality; abstract behind an interface either way                   |
| OQ-5  | Payments: do we hold funds (escrow-like) or route directly to the owner with commission invoiced separately? Material legal and regulatory implications                   | Founder + counsel        | Month 8                         | Direct routing with separate commission invoice                                         |
| OQ-6  | Data policy for aggregated benchmarks: opt-in default, contractual terms and anonymisation standard                                                                       | Founder + counsel        | Before V2.0                     | Strict opt-in, k-anonymised, never at individual board level                            |
| OQ-7  | Rate visibility default on the marketplace: card rate shown, band shown, or on request?                                                                                   | Product Owner + GTM      | Before V2.0                     | Card rate shown; owner may override per board                                           |
| OQ-8  | Brand architecture: keep HOARDINGS360 as master brand, or adopt a broader parent brand?                                                                                   | Founder + Marketing      | Sprint 0                        | Keep HOARDINGS360; revisit at V4.0                                                      |
| OQ-9  | Do we serve advertisers directly with a managed planning service in early releases (high conversion, non-scalable), or stay pure platform?                                | Founder                  | V2.0                            | Yes, manually, and instrument it — it is the fastest route to learning what buyers want |
| OQ-10 | Native mobile apps: needed at V4.0, or does the PWA suffice?                                                                                                              | Product Owner + Eng Lead | V3.0                            | PWA; reassess against field feedback                                                    |
| OQ-11 | Commission introduction: how do we transition existing owners onto a take-rate without churn?                                                                             | GTM Lead                 | Before V3.0                     | 60-day notice; marketplace-originated only; grandfather Founding Partners for 12 months |
| OQ-12 | Do we support tenants who manage boards on behalf of third-party owners (sub-ownership)? Affects the boards and leases schema                                             | Product Owner            | <b>Sprint 0</b> (schema impact) | Yes — model `ownership_type` and a managing-tenant relationship from day one            |

## 23.2 Assumptions this plan rests on

28. Media owners in the target segment have a real, felt, recurring pain around permit expiry and vacancy — strong enough to pay for. **Validated qualitatively; to be validated commercially by V1.0 exit criterion 6.**
29. Owners will permit their inventory to be published publicly with availability dates. **Test explicitly with the first 10 partners; if a material share refuse, the marketplace thesis needs rework (see OQ-7).**
30. Advertiser demand can be acquired primarily through organic search at acceptable cost. **Validated by V2.0 exit criteria; if not, the plan shifts weight to agency partnerships.**
31. A field digitisation squad can capture a board in under 15 minutes at a cost that makes onboarding economics work. **Measure from the first week of the census.**
32. The team can be staffed per Section 22.1 within the stated windows.
33. No dominant competitor achieves state-level inventory density before we do.

## 23.3 What would make us change the plan

- If fewer than 6 of 10 design partners use the product weekly at V1.0 exit → **stop; the wedge is wrong.** Re-interview and re-scope before building the marketplace.
- If owners refuse public availability publication at scale → pivot to a private, invitation-based inventory exchange for agencies rather than a public marketplace.
- If organic search cannot deliver qualified enquiries at acceptable cost by V2.0 exit → shift demand strategy to agency and enterprise buyer relationships, and slow SEO investment.
- If leakage exceeds 60% at V3.0 → abandon commission as a primary stream and reprice subscription upward, positioning the marketplace as a subscription benefit rather than a transaction channel.

## Appendix A — Glossary

| Term                   | Meaning                                                                                       |
|------------------------|-----------------------------------------------------------------------------------------------|
| OOH / DOOH             | Out-of-Home advertising / Digital Out-of-Home (screen-based)                                  |
| Board / structure      | A physical advertising asset (hoarding, unipole, gantry, LED screen, shelter)                 |
| Face                   | An individually sellable advertising surface on a structure; one structure may have several   |
| Card rate              | The published monthly rental rate for a face, before negotiation                              |
| Floor rate             | The minimum rate a sales executive may accept; internal only                                  |
| Occupancy              | Whether a face is sold for a given period — distinct from lifecycle and compliance status     |
| Vacancy loss           | Revenue foregone while a face is unsold, valued at card rate                                  |
| Pre-listing window     | The period before a contract ends during which the face is published as upcoming availability |
| Proof of Display / PoP | Geo-tagged, timestamped photographic evidence that a campaign is installed and live           |
| NOC                    | No Objection Certificate — a clearance issued by an authority such as traffic police          |
| Landowner lease        | The agreement under which the structure occupies someone else's land, with rent payable       |
| Tenant                 | One media owner company on the platform; the data isolation boundary                          |
| Leakage                | Demand matched on the platform that transacts off it                                          |
| GMV                    | Gross Merchandise Value — total campaign value booked through the platform                    |
| DPDP Act               | India's Digital Personal Data Protection Act, 2023                                            |
| DLT                    | Distributed Ledger Technology registration required for commercial SMS in India               |
| BSP                    | Business Solution Provider — an authorised WhatsApp Business API partner                      |

## Appendix B — Notification Matrix (V1.0–V1.1)

| Trigger               | Timing                               | Recipients                            | Channel                             | Priority        |
|-----------------------|--------------------------------------|---------------------------------------|-------------------------------------|-----------------|
| Permit expiring       | 90 / 60 / 30 / 15 / 7 days before    | Compliance officer, company admin     | Email + in-app (WhatsApp from V1.1) | High            |
| Permit expired        | On expiry, then daily until resolved | Compliance officer, company admin     | WhatsApp + email + in-app           | <b>Critical</b> |
| Agreement expiring    | 90 / 60 / 30 / 15 / 7 days before    | Assigned sales executive, ops manager | Email + in-app (WhatsApp from V1.1) | High            |
| Board becoming vacant | At the pre-listing window            | Sales team, ops manager               | WhatsApp + in-app                   | High            |
| Board vacant          | On the day it becomes vacant         | Sales team                            | In-app + daily digest               | Medium          |
| Import completed      | On completion                        | Uploading user                        | Email + in-app                      | Low             |
| Incident created      | Immediate                            | Ops manager, supervisor,              | WhatsApp + in-app                   | <b>Critical</b> |

| Trigger                      | Timing                     | Recipients                               | Channel                                  | Priority        |
|------------------------------|----------------------------|------------------------------------------|------------------------------------------|-----------------|
| (high/critical)              |                            | account manager if occupied              |                                          |                 |
| Incident unresolved          | At SLA breach (V4.0)       | Supervisor, then ops manager             | WhatsApp + email                         | High            |
| Proof uploaded               | On upload                  | Account manager (client from V3.0)       | In-app (email to client)                 | Medium          |
| Enquiry received (V2.0)      | Immediate                  | Sales team, company admin                | WhatsApp + email + in-app                | High            |
| Enquiry unanswered (V2.0)    | 24 hours                   | Company admin; platform analytics flag   | Email + in-app                           | High            |
| Hold expiring (V2.0)         | 6 hours before expiry      | Advertiser and owner                     | Email + in-app                           | Medium          |
| Landowner payment due (V4.0) | 15 / 7 / 1 days before     | Finance officer, company admin           | WhatsApp + email                         | <b>Critical</b> |
| Invoice overdue (V3.0)       | 7 / 15 / 30 days after due | Finance officer, account manager, client | Email (+ WhatsApp to client if opted in) | High            |

## Appendix C — Illustrative Schema Extracts

Indicative only — the Engineering Lead owns the final schema. Included to make the guardrails in Section 8 concrete.

```
-- Structures and their independently sellable faces
CREATE TABLE boards (
  board_id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id         UUID NOT NULL REFERENCES tenants(tenant_id),
  board_code        VARCHAR(40) NOT NULL,          -- human-readable, e.g. BLR-HSR-0142
  name              VARCHAR(200) NOT NULL,
  structure_type     VARCHAR(50) NOT NULL,          -- configurable taxonomy
  location           geography(Point,4326) NOT NULL,
  address_line       VARCHAR(300), landmark VARCHAR(200),
  city              VARCHAR(100), district VARCHAR(100), state VARCHAR(100),
  ward_zone          VARCHAR(100), road_name VARCHAR(200), road_category VARCHAR(50),
  ownership_type     VARCHAR(30) NOT NULL,          -- owned | leased_in |
  managed_for_third_party
  installed_on       DATE,
  lifecycle_status   VARCHAR(30) NOT NULL DEFAULT 'draft',
  last_verified_at   TIMESTAMPTZ,
  created_at         TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  deleted_at         TIMESTAMPTZ, deleted_by UUID, deletion_reason TEXT,
  UNIQUE (tenant_id, board_code)
);
CREATE INDEX idx_boards_location ON boards USING GIST (location);
ALTER TABLE boards ENABLE ROW LEVEL SECURITY;

CREATE TABLE board_faces (
  face_id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id          UUID NOT NULL,
  board_id           UUID NOT NULL REFERENCES boards(board_id),
  face_label         VARCHAR(20) NOT NULL,          -- A, B, North, South
  facing_direction   VARCHAR(30),
  width_ft           NUMERIC(6,2), height_ft NUMERIC(6,2),
```



```

illumination    VARCHAR(30),                -- frontlit | backlit | nonlit | led
visibility_m    INT,
card_rate_paise BIGINT, floor_rate_paise BIGINT,
is_publishable BOOLEAN NOT NULL DEFAULT FALSE,
UNIQUE (board_id, face_label)
);

-- Compliance: MANY independent clearances per board, each with its own cycle
CREATE TABLE compliance_records (
  compliance_id  UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id      UUID NOT NULL,
  board_id       UUID NOT NULL REFERENCES boards(board_id),
  clearance_type VARCHAR(60) NOT NULL,    -- configurable per tenant
  governing_body VARCHAR(200) NOT NULL,
  reference_no   VARCHAR(100),
  issue_date DATE, expiry_date DATE,
  renewal_cycle_months INT,
  fee_paid_paise BIGINT,
  status         VARCHAR(30) NOT NULL DEFAULT 'valid', -- COMPUTED, never set by hand
  supersedes_id  UUID REFERENCES compliance_records(compliance_id),
  created_at     TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
CREATE INDEX idx_compliance_expiry ON compliance_records (tenant_id, expiry_date)
  WHERE status <> 'superseded';

-- One polymorphic repository for every document in the business
CREATE TABLE documents (
  document_id  UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id    UUID NOT NULL,
  entity_type   VARCHAR(50) NOT NULL, -- board | campaign | agreement | permit |
                                     -- incident | work_order | lease | vendor | client
  entity_id     UUID NOT NULL,
  document_type VARCHAR(60) NOT NULL,
  issuing_authority VARCHAR(200),
  reference_no   VARCHAR(100),
  issue_date DATE, expiry_date DATE,
  version_no     INT NOT NULL DEFAULT 1,
  supersedes_id  UUID REFERENCES documents(document_id),
  signature_status VARCHAR(30) NOT NULL DEFAULT 'not_required',
  storage_key     VARCHAR(500) NOT NULL, checksum VARCHAR(128),
  uploaded_by    UUID NOT NULL, uploaded_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  deleted_at     TIMESTAMPTZ
);
CREATE INDEX idx_documents_entity ON documents (tenant_id, entity_type, entity_id);

-- Derived occupancy: the engine behind availability and the vacancy loop
CREATE TABLE occupancy_periods (
  period_id  UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id  UUID NOT NULL,
  face_id    UUID NOT NULL REFERENCES board_faces(face_id),
  start_date DATE NOT NULL, end_date DATE NOT NULL,
  state      VARCHAR(30) NOT NULL, -- occupied | on_hold | blocked | booked_future
  source_type VARCHAR(30) NOT NULL, -- agreement | hold | manual_block
  source_id   UUID,
  EXCLUDE USING GIST ( -- prevents double-booking at the database level
    face_id WITH =,
    daterange(start_date, end_date, '[]') WITH &&
  ) WHERE (state IN ('occupied', 'booked_future'))
);

```

```
-- Universal, append-only activity feed
CREATE TABLE activity_events (
  event_id    BIGSERIAL PRIMARY KEY,
  tenant_id   UUID NOT NULL,
  event_type  VARCHAR(60) NOT NULL,
  entity_type VARCHAR(50) NOT NULL, entity_id UUID NOT NULL,
  actor_id    UUID, actor_type VARCHAR(30) NOT NULL DEFAULT 'user',
  payload     JSONB NOT NULL,
  schema_version INT NOT NULL DEFAULT 1,
  occurred_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
CREATE INDEX idx_activity_entity ON activity_events (tenant_id, entity_type, entity_id,
occurred_at DESC);
```

## Appendix D — Onboarding Import Specification (V1.0)

The columns below define the customer-facing import template. Owners will submit messy files; the importer must be forgiving on format and strict on meaning.

| Sheet      | Required columns                                                   | Optional columns                                                                          | Validation                                                                                                                |
|------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Boards     | board_code, name, structure_type, latitude, longitude, city, state | address, landmark, district, ward, road_name, road_category, ownership_type, installed_on | Coordinates within India and plausible for the stated city; duplicate detection within 25 m; board_code unique per tenant |
| Faces      | board_code, face_label, width_ft, height_ft, illumination          | facing_direction, visibility_m, card_rate, floor_rate                                     | Referenced board_code must exist; dimensions positive; at least one face per board                                        |
| Clients    | client_name                                                        | gst_no, billing_address, contact_name, phone, email, payment_terms                        | GST format check where supplied; duplicate name warning                                                                   |
| Agreements | board_code, face_label, client_name, start_date, end_date, value   | agreement_ref, payment_schedule, notes                                                    | End date after start date; no overlapping agreement on the same face; client must exist or be created                     |
| Permits    | board_code, clearance_type, governing_body, expiry_date            | reference_no, issue_date, renewal_cycle_months, fee_paid                                  | Expiry date present and parseable; past expiry accepted but flagged as expired                                            |

- Accept DD-MM-YYYY, DD/MM/YYYY and Excel serial dates. Reject ambiguous dates with a clear message rather than guessing.
- Accept rates written as "45000", "45,000", "₹45,000" and "45k".
- Accept coordinates as decimal degrees or as a pasted Google Maps link, and parse both.
- Return a correction file containing only the failed rows, with an error column appended — never make the owner re-upload the whole file.

## Appendix E — Marketplace Launch Checklist (V2.0)

| Area      | Item                                                                         | Owner            |
|-----------|------------------------------------------------------------------------------|------------------|
| Inventory | ≥ 3,000 published faces from ≥ 25 verified owners                            | GTM Lead         |
| Inventory | ≥ 85% of faces with photo < 12 months, GPS verified, rate set, permits valid | Customer Success |

| Area        | Item                                                                                    | Owner            |
|-------------|-----------------------------------------------------------------------------------------|------------------|
| Inventory   | Duplicate structures across tenants resolved                                            | Platform Ops     |
| Legal       | Terms of use, privacy notice, listing policy, neutrality policy published               | Legal / Founder  |
| Legal       | DPDP compliance review completed; consent flows implemented                             | Legal / Eng Lead |
| Security    | Third-party penetration test complete, critical and high findings closed                | Engineering Lead |
| Security    | Tenant-isolation test suite green; rate limiting and bot protection live                | QA               |
| Performance | Load test at 10× projected launch traffic passed; CDN and caching verified              | DevOps           |
| SEO         | ≥ 40 landing pages published, indexed, sitemap submitted, Core Web Vitals within budget | Marketing        |
| Operations  | Every listed owner has committed in writing to the 24-hour enquiry SLA                  | Customer Success |
| Operations  | Support rota, runbook and escalation path published                                     | Product Owner    |
| Analytics   | Full funnel instrumentation verified end to end in production                           | Product Owner    |
| Marketing   | PR, launch event, demo video, live inventory counter, founding partner logos ready      | Marketing        |
| Contingency | Rollback plan and "unpublish everything" switch tested                                  | Engineering Lead |

## Appendix F — Document Control

| Version | Date     | Author        | Change                                                                                                                                                                                                                                                                        |
|---------|----------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v1.0    | Baseline | Product Owner | Consolidates the original OOH_Marketplace_PRD, the HOARDINGS360 validation document, the 10-module operational analysis and the enterprise architecture recommendations into a single master document, adding release sequencing, go-to-market, metrics, risk and governance. |

**Change control:** any change to scope, sequencing or architecture guardrails requires a new version of this document, approved by the Product Owner and the Engineering Lead, and recorded above. Day-to-day decisions are recorded in the decision log (Section 22.2), which is reconciled into this document at every release re-baseline.

### Closing note to the team

Everything in this document is subordinate to one idea: **the fastest path to a valuable marketplace runs through being genuinely useful to media owners on an ordinary Tuesday morning.** Compliance alerts, occupancy dashboards, proposal builders and proof photographs are not stepping stones to the real product — for the first year, they **are** the product. The marketplace is what that trust earns us.